

Econometrics of Commodity and Asset Pricing

Lecture 2

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- I. Forwards, Futures, Dividends, Commodity Pricing and Convenience Yields
- II. Modelling Spot and Forward Electricity Prices
- III. Simultaneous Modelling of Several Commodity Markets
- IV. Basic Concepts in Fixed Income Analysis

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Chapter I

FORWARDS, FUTURES, DIVIDENDS, COMMODITY PRICING AND CONVENIENCE YIELDS

– Agenda –

- I.1 Forward contracts, forward prices. (Slide 4)
- I.2 Futures contracts, futures prices. (Slide 12)
- I.3 Convenience yields. (Slide 16)
- I.4 Pricing with affine models. (Slide 19)
- I.5 A Gaussian VAR model. (Slide 21)
- I.6 Applications. (Slide 25)

I.1 – FORWARD CONTRACTS, FORWARD PRICES

□ This chapter introduces Forwards and Futures contracts.

- We begin by establishing core definitions, differentiating these contracts based on their cash flow mechanics and settlement procedures.
- We then address the pricing problem, deriving fundamental relationships within a discrete-time framework.
- We subsequently demonstrate how affine processes yield tractable pricing formulas, even when underlying assets and stochastic discount factors exhibit complex dynamics.

□ This analysis is extended to incorporate dividend-paying assets:

- we modify the pricing equations to account for associated cash flows, which is particularly relevant for equities and fixed-income instruments.

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I.1 – FORWARD CONTRACTS, FORWARD PRICES

- A significant portion of the chapter focuses on pricing commodity derivatives.
 - We recognize the distinct characteristics of storable commodities and we introduce the concept of **convenience yield**.
 - The convenience yield represents the net benefit of physically holding a commodity, encompassing both advantages (e.g., meeting unexpected demand) and disadvantages (e.g., storage costs).
 - We demonstrate how to adapt the previously derived pricing results to commodity markets using this concept.
- We end this chapter by quickly presenting the work of Schwartz (1997) on the stochastic behavior of commodity prices.

I.1 – FORWARD CONTRACTS, FORWARD PRICES

□ Notation:

- w_t denotes the information at t ;
- S_t is value of the asset (commodity) at t , (function of w_t);
- r_t is the risk-free short rate between t and $t + 1$ (known at t), (function of w_t);
- $B(t_1, t_2 - t_1)$: date- t_1 price of a zero-coupon bond whose value is 1 on t_2 .

□ **Forward contract:** it is an agreement signed at t (today) to buy or sell an asset (a commodity) at a given delivery date $T > t$ at a specified price K (**called delivery price**) decided at t .

□ Since a Forward contract is settled at maturity and a party in the long (short) position is obliged to buy (to sell) the asset being worth S_T at maturity for K , the **payoff** of the long (short) position is the **contingent claim** $g_T = S_T - K$ ($-g_T$).

□ Observe that there is no cash flow at t and, therefore, the **price of a forward contract at its initial date is zero**.

I.1 – FORWARD CONTRACTS, FORWARD PRICES

- The **delivery price** K which induces a **worthless forward contract** at t is called the **forward price** (of the underlying asset valued S_T for the settlement date T) and is denoted $\Phi_{t,T}$.
- For all **initial dates** $t < T$, the forward price will be denoted $\Phi_{t,T}$ and, for $t = T$ (immediate delivery), we have by definition $\Phi_{T,T} := S_T$.

Prop. I.1 (Forward price)

The forward price $\Phi_{t,T}$ at date t is such that $0 = \mathbb{E}_t [M_{t,T} (S_T - \Phi_{t,T})]$ and, therefore, it is given by:

$$\Phi_{t,T} = \frac{\mathbb{E}_t(M_{t,t+1}, \dots, M_{T-1,T} S_T)}{B(t, T-t)} = \frac{\mathbb{E}_t^Q[\exp(-r_t - \dots - r_{T-1}) S_T]}{B(t, T-t)}.$$

and it satisfies the recursive relation :

$$\begin{aligned}\Phi_{t,T} &= \mathbb{E}_t \left[\frac{M_{t,t+1}}{B(t, T-t)} B(t+1, T-t-1) \Phi_{t+1,T} \right] \\ &= \mathbb{E}_t^Q \left[\frac{B(t, 1) B(t+1, T-t-1)}{B(t, T-t)} \Phi_{t+1,T} \right],\end{aligned}$$

where $B(t, T-t)$ is the price, at date t , of a ZCB maturing at time T .

I.1 – FORWARD CONTRACTS, FORWARD PRICES

- **Proof:** Because the price of the forward contract at t is zero we have, under the no-arbitrage principle, that :

$$0 = \mathbb{E}_t \left\{ M_{t,t+1} \cdot \dots \cdot M_{T-1,T} \times [S_T - \Phi_{t,T}] \right\} .$$

Solving for the forward price we obtain :

$$\begin{aligned} \Phi_{t,T} &= \frac{\mathbb{E}_t \left[M_{t,t+1} \cdot \dots \cdot M_{T-1,T} S_T \right]}{B(t, T - t)} \\ &= \frac{\mathbb{E}_t \left\{ M_{t,t+1} B(t+1, T - t - 1) \frac{\mathbb{E}_{t+1} [M_{t+1,t+2} \cdot \dots \cdot M_{T-1,T} S_T]}{B(t+1, T - t - 1)} \right\}}{B(t, T - t)} \\ &= \mathbb{E}_t \left[M_{t,t+1} \frac{B(t+1, T - t - 1)}{B(t, T - t)} \Phi_{t+1,T} \right] . \end{aligned}$$

I.1 – FORWARD CONTRACTS, FORWARD PRICES

Remark I.1

If the processes (r_t) and (S_t) are $\mathbb{Q}$ -independent, then

$$\Phi_{t,T} = \mathbb{E}_t^{\mathbb{Q}}(S_T)$$

and $\Phi_{t,T}$ is a $\mathbb{Q}$ -martingale.

Remark I.2

If the asset does not generate any payoff before T (no dividends), then

$$\Phi_{t,T} = \frac{S_t}{B(t, T - t)}.$$

Remark I.3

The price at s , with $t < s < T$, of a forward contract signed at t is:

$$\mathbb{E}_s[M_{s,T} S_T] - \Phi_{t,T} B(s, T - s) = (\Phi_{s,T} - \Phi_{t,T}) B(s, T - s),$$

that is, the payoff $S_T - \Phi_{t,T}$ at T has this price at s .

I.1 – FORWARD CONTRACTS, FORWARD PRICES

□ Let us introduce now **dividend yield and payment**. Notations:

- $\tilde{S}_t$: ex-dividend price at t ;
- S_t : cum-dividend price at t ;
- $S_t = \tilde{S}_t \exp(\delta_t)$, δ_t : dividend yield (or rate) known at t ;
- thus, the dividend payment is defined as $\tilde{S}_t(\exp(\delta_t) - 1)$.

□ We get:

$$\tilde{S}_{T-1} = S_{T-1} \exp(-\delta_{T-1}) = \mathbb{E}_{T-1}(M_{T-1,T} S_T)$$

or

$$\begin{aligned} S_{T-1} &= \mathbb{E}_{T-1}[M_{T-1,T} \exp(\delta_{T-1}) S_T], \\ &= \mathbb{E}_{T-1}^Q[\exp(-r_{T-1}) \exp(\delta_{T-1}) S_T] \end{aligned}$$

and, recursively:

$$\begin{aligned} S_t &= \mathbb{E}_t[M_{t,t+1} \dots M_{T-1,T} \exp(\delta_t + \dots + \delta_{T-1}) S_T] \\ &= \mathbb{E}_t^Q[\exp(-r_t \dots - r_{T-1} + \delta_t + \dots + \delta_{T-1}) S_T]. \end{aligned}$$

$\Rightarrow r_t$ replaced by $r_t - \delta_t$.

1.1 – FORWARD CONTRACTS, FORWARD PRICES

Remark 1.4

Replacing S_t by $\tilde{S}_t \exp(\delta_t)$ and S_T by $\tilde{S}_T \exp(\delta_T)$, we get

$$\tilde{S}_t = \mathbb{E}_t^{\mathbb{Q}} \left[\exp(-r_t - \dots - r_{T-1} + \delta_{t+1} + \dots + \delta_T) \tilde{S}_T \right].$$

Remark 1.5

If the dividend yield δ_t is deterministic, then from Proposition 1.1 and the cum-dividend pricing formula we obtain:

$$\Phi_{t,T} = \frac{S_t \exp(-\delta_t - \dots - \delta_{T-1})}{B(t, T - t)}.$$

If r_t is deterministic too, then we have:

$$\Phi_{t,T} = S_t \exp(r_t + \dots + r_{T-1} - \delta_t - \dots - \delta_{T-1}).$$

Here, the negative signs preceding the dividend yields capture the fact that when the asset is purchased forward, the buyer forfeits the dividend payments made between dates t and T .

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I.2 – FUTURES CONTRACTS, FUTURES PRICES

- A **Futures contract** is an agreement, signed at the initial date t , to buy or sell a **standardized** asset (commodity) at a given future date T , and for a given price $F_{t,T}$ decided at t and called the **Futures price**.

- Futures contracts are similar to Forward contracts, yet there are some differences:
 - They are traded on a *regulated exchange*, such as **Chicago Board of Trade (CBOT)** or the **Chicago Mercantile Exchange (CME)**, that defines the characteristics of the contract, and it guarantees that the payments will be honored at maturity.
 - The security underlying the contract is **standardized**, in the sense that the future contract *clearly specifies* the type of security this is eligible for delivery, as well as the time and the method of delivery of the security.

I.2 – FUTURES CONTRACTS, FUTURES PRICES

□ **Futures contracts** features the "**marking to market**" procedure:

- the Futures contract is worth zero at t (inception of the contract), and then both counterparties are required at **every trading day** $s > t$ to deposit into a **margin account** a resettlement payment (**margin call**).
- In order to enter into a Futures position, a trader must post an initial amount of money (**initial margin**) in the margin account. As the Futures price moves, the margin account gets debited or credited, depending on the movement.
- If we consider the Futures price process $\mathbf{F} = \{F_{t,T}, \dots, F_{T,T}\}$, at the end of each trading day s (with $t < s \leq T$), the margin call for the buyer (balance of the investor's margin account) is the amount $\Delta_{s,T} = F_{s,T} - F_{s-1,T}$.
- In other words, Futures contracts are closed out after each trading day, and then start afresh the next trading day. Its value is therefore zero at every date s .

I.2 – FUTURES CONTRACTS, FUTURES PRICES

Prop. 1.2

The Futures price at date t is given by :

$$F_{t,T} = \mathbb{E}_t^{\mathbb{Q}} [S_T] ,$$

and satisfies the recursive relation :

$$F_{t,T} = \mathbb{E}_t^{\mathbb{Q}} [F_{t+1,T}] .$$

So the Futures Price process $\{F_{s,T}\}_{t \leq s \leq T}$ is a $\mathbb{Q}$ -martingale or, in other words, the process $(\Delta_{s,T})_{t < s \leq T}$ is a $\mathbb{Q}$ -martingale difference.

Proof:

- At each date $s > t$, after the deposit of the resettlement payment, there is a new contract valued zero and paying $F_{s+1,T} - F_{s,T}$ at $s + 1$ (and providing another zero valued contract at $s + 1$).
- Therefore $0 = \mathbb{E}_s^{\mathbb{Q}}[\exp(-r_s)(F_{s+1,T} - F_{s,T})]$, and $\mathbb{E}_s^{\mathbb{Q}}(F_{s+1,T} - F_{s,T}) = 0$, since $\exp(-r_s)$ is known at s .
- Hence $F_{s,T} = \mathbb{E}_s^{\mathbb{Q}}(F_{s+1,T})$, and the results follows from $F_{T,T} = S_T$. ■

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I.2 – FUTURES CONTRACTS, FUTURES PRICES

Prop. 1.3

$$\Phi_{t,T} - F_{t,T} = \frac{\text{cov}_t^{\mathbb{Q}} \left[\prod_{s=t}^{T-1} \exp(-r_s), S_T \right]}{B(t, T-t)}.$$

Proof.

$$\begin{aligned}\Phi_{t,T} - F_{t,T} &= \mathbb{E}_t^{\mathbb{Q}} \left[\frac{\prod_{s=t}^{T-1} \exp(-r_s) S_T}{B(t, T-t)} - S_T \right] \\&= \frac{\mathbb{E}_t^{\mathbb{Q}} \left[\prod_{s=t}^{T-1} \exp(-r_s) S_T \right] - \mathbb{E}_t^{\mathbb{Q}} \left[\prod_{s=t}^{T-1} \exp(-r_s) \right] \mathbb{E}_t^{\mathbb{Q}}(S_T)}{B(t, T-t)} \\&= \frac{\text{cov}_t^{\mathbb{Q}} \left[\prod_{s=t}^{T-1} \exp(-r_s), S_T \right]}{B(t, T-t)}. \blacksquare\end{aligned}$$

Remark 1.6

$\Phi_{t,T} = F_{t,T}$, if, and only if, $\prod_{s=t}^{T-1} \exp(-r_s)$ and S_T are conditionally uncorrelated under $\mathbb{Q}$. In particular it is true in the case of deterministic short rates.

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I.3 – COMMODITY PRICING AND CONVENIENCE YIELDS

Definition I.1 (Convenience yield)

A **convenience yield** is a net benefit associated with holding a physical asset (rather than a forward or futures contract on that asset). It is "net" in the sense that it is equal to the positive gain of holding minus the potential storage costs.

- The notion of convenience yield is valid only for storable commodities (not, e.g., for electricity).
 - It can be positive or negative.
 - The notion of convenience yield is mathematically similar to a dividend yield (but can be also < 0 , and it is often unobservable).
- In the following, we denote the convenience yield by c_t .
 - The price S_t is here the "cum" convenience yield price. It is the price at which one unit of the asset can be purchased on financial markets.
 - Given our convention regarding the definition of the convenience yield, we have for instance that $c_t < 0$ if there is no gain and only storage costs.

Basic pricing approach

- r_t and c_t deterministic. We have:

$$\Phi_{t,T} = F_{t,T} = S_t \exp(r_t + \dots + r_{T-1} - c_t - \dots - c_{T-1}).$$

- Moreover, if r_t and c_t time independent:

$$\Phi_{t,T} = F_{t,T} = S_t \exp[(T - t)(r - c)].$$

- If $r > c$ forward curve function $\uparrow$ of T : **situation of "contango"**;
- If $r < c$ forward curve function $\downarrow$ of T : **"backwardated" forward curve**.

Advanced approach: c_t is stochastic.

- We have:

$$\Phi_{t,T} = \mathbb{E}_t^{\mathbb{Q}}[\exp(-r_t - \dots - r_{T-1})S_T]/B(t, T - t) \quad (I.1)$$

$$F_{t,T} = \mathbb{E}_t^{\mathbb{Q}}(S_T) \quad (I.2)$$

$$S_t = \mathbb{E}_t^{\mathbb{Q}}[\exp(-r_t - \dots - r_{T-1} + c_t + \dots + c_{T-1})S_T]. \quad (I.3)$$

- necessitates a joint modelling (at least under $\mathbb{Q}$) of: S_t (or $s_t = \log S_t$), c_t (convenience yield), r_t (short rate),
- and, in general, other factors (observable or latent). Direct or Back Modelling approach (see, for instance, Section I.4).

Remark 1.7 (Case of a storable commodity)

Given that $S_t = \mathbb{E}_t^{\mathbb{Q}}[\exp(-r_t + c_t)S_{t+1}]$ (storage costs $c_t < 0$ imply a smaller price S_t), then the date- t ex-convenience-yield price, i.e. $S_t \exp(-c_t)$, is the price at t of S_{t+1} :

$$\begin{aligned} S_t \exp(-c_t) &= \exp(-r_t) \mathbb{E}_t^{\mathbb{Q}}(S_{t+1}) \\ \mathbb{E}_t^{\mathbb{Q}}(S_{t+1}) &= S_t \exp(r_t - c_t) \\ \mathbb{E}_t^{\mathbb{Q}} \exp(s_{t+1}) &= \exp(s_t + r_t - c_t). \end{aligned} \tag{1.4}$$

If $y_{t+1} = \log \frac{S_{t+1}}{S_t}$ is the return, then $\mathbb{E}_t^{\mathbb{Q}} \exp(y_{t+1}) = \exp(r_t - c_t)$. Eq. (1.3) can be obtained by using Eq. (1.4) recursively, and the latter implies ICCs.

Remark 1.8 (Case of a non-storable commodity)

There is no reason to assume that the price at date t is equal to the discounted value of its future price, as it is not possible to purchase the commodity, store it and sell it at a later date (like in a typical (no-)arbitrage strategy). Given $c_t = 0$ and if (in addition) r_t is deterministic, then:

$$\Phi_{t,T} = F_{t,T} = \mathbb{E}_t^{\mathbb{Q}}(S_T),$$

But, since Eq. (1.3) is not valid we do not have $S_t = B(t, T) \mathbb{E}_t^{\mathbb{Q}}(S_T)$ and therefore we do not have $\Phi_{t,T} = \mathbb{E}_t^{\mathbb{Q}}(S_T) = S_t / B(t, T)$.

I.4 – PRICING WITH AFFINE MODELS

Risk neutral dynamics (Back Modelling)

- We have a storable commodity:

$$w_t = (s_t, c_t, r_t, x_t')',$$

where $s_t = \log S_t$ and x_t is a vector of additional factors. w_t is $\mathbb{Q}$ -affine:

$$\mathbb{E}_t^{\mathbb{Q}} \exp(u' w_{t+1}) = \exp[a'(u) w_t + b(u)].$$

- If $S_{t+1} = \exp(e_1' w_{t+1})$, ICCs give:

$$\begin{aligned} \mathbb{E}_t^{\mathbb{Q}} \exp(s_{t+1}) &= \exp(s_t - c_t + r_t) \\ \Leftrightarrow \mathbb{E}_t^{\mathbb{Q}} \exp(e_1' w_{t+1}) &= \exp[(e_1 - e_2 + e_3)' w_t] \\ \Leftrightarrow \begin{cases} a(e_1) &= e_1 - e_2 + e_3 \\ b(e_1) &= 0 \end{cases} \end{aligned}$$

(if non storable, no c_t , no ICCs). Since (Eqs. I.1 and I.2):

$$\begin{aligned} \Phi_{t,T} &= \frac{\mathbb{E}_t^{\mathbb{Q}}[\exp(-r_t - \dots - r_{T-1} + s_T)]}{\mathbb{E}_t^{\mathbb{Q}}[\exp(-r_t - \dots - r_{T-1})]} \\ F_{t,T} &= \mathbb{E}_t^{\mathbb{Q}}[\exp(s_T)], \end{aligned}$$

$\Rightarrow$ quasi explicit formulas using multihorizon Laplace transforms, exponential affine in $w_t = (s_t, c_t, r_t, x_t')'$.

I.4 – PRICING WITH AFFINE MODELS

Historical dynamics

□ We have:

$$\mathcal{M}_{t,t+1}(\underline{w}_{t+1}) = \exp[-r_t + \alpha_t' w_{t+1} + \psi_t^{\mathbb{Q}}(-\alpha_t)],$$

and

$$\mathbb{E}_t(\mathcal{M}_{t,t+1}) = \exp(-r_t).$$

- Historical dynamics defined through:

$$\psi_t(u) = \psi_t^{\mathbb{Q}}(u - \alpha_t) - \psi_t^{\mathbb{Q}}(-\alpha_t).$$

- Note: no constraint on α_t , possibility to introduce seasonality.

I.5 – A GAUSSIAN VAR MODEL

□ State vector: $w_t = (s_t, c_t, r_t)'$.

- R.N. dynamics:

$$w_{t+1} = A_0 + A_1 w_t + \varepsilon_{t+1}, \quad \varepsilon_{t+1} \sim i.i.d. \mathcal{N}(0, \Sigma) \text{ under } \mathbb{Q}.$$

- We have (see Lecture 1):

$$\begin{aligned} \mathbb{E}_t^{\mathbb{Q}} \exp(u' w_{t+1}) &= \exp \left[u' (A_0 + A_1 w_t) + \frac{1}{2} u' \Sigma u \right] \\ \Rightarrow \quad \begin{cases} a(u) &= A_1' u \\ b(u) &= A_0' u + \frac{1}{2} u' \Sigma u. \end{cases} \end{aligned}$$

- Let us decompose A_0 and A_1 as follows:

$$A_0 = \begin{pmatrix} A_{01} \\ \tilde{A}_0 \end{pmatrix}, \quad A_1 = \begin{pmatrix} A_{11} \\ \tilde{A}_1 \end{pmatrix}.$$

- ICCs (if storable, otherwise no constraint):

$$\begin{cases} a(e_1) &= e_1 - e_2 + e_3 \\ b(e_1) &= 0. \end{cases}$$

$\Leftrightarrow$ The first row of A_1 is $A_{11} = e'_1 - e'_2 + e'_3$, the first element of A_0 is $A_{01} = -\frac{1}{2}\sigma_1^2$ (σ_1^2 conditional variance of s_{t+1}).

- In other words the $\mathbb{Q}$ -VAR is:

$$\begin{cases} s_{t+1} &= -\frac{1}{2}\sigma_1^2 + s_t - c_t + r_t &+ \varepsilon_{1,t+1} \\ \begin{pmatrix} c_{t+1} \\ r_{t+1} \end{pmatrix} &= \tilde{A}_0 + \tilde{A}_1 w_t &+ \begin{pmatrix} \varepsilon_{2,t+1} \\ \varepsilon_{3,t+1} \end{pmatrix}, \end{cases}$$

(where $\tilde{A}_0$ and $\tilde{A}_1$ are not constrained).

- Given $y_{t+1} = \log(S_{t+1}/S_t) = s_{t+1} - s_t$, we have:

$$\mathbb{E}_t^{\mathbb{Q}}(y_{t+1}) = r_t - c_t - \frac{1}{2}\sigma_1^2,$$

$\Rightarrow y_{t+1} - r_t + c_t + \frac{1}{2}\sigma_1^2$ is a $\mathbb{Q}$ -martingale difference.

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- Historical dynamics:

$$\begin{aligned}
 \psi_t(u) &= \psi_t^{\mathbb{Q}}(u - \alpha_t) - \psi_t^{\mathbb{Q}}(-\alpha_t) \\
 &= u'(A_0 + A_1 w_t) + \frac{1}{2}(u - \alpha_t)' \Sigma (u - \alpha_t) - \frac{1}{2} \alpha_t' \Sigma \alpha_t \\
 &= u'(A_0 + A_1 w_t) - u' \Sigma \alpha_t + \frac{1}{2} u' \Sigma u.
 \end{aligned}$$

- If we take $\alpha_t = \alpha_0 + \alpha_1 w_t$, we get:

$$\begin{aligned}
 \psi_t(u) &= u'[A_0 - \Sigma \alpha_0 + (A_1 - \Sigma \alpha_1) w_t] + \frac{1}{2} u' \Sigma u \\
 \Rightarrow w_{t+1} &= (A_0 - \Sigma \alpha_0) + (A_1 - \Sigma \alpha_1) w_t + \xi_t, \\
 \xi_t &\sim i.i.d. \mathcal{N}(0, \Sigma) \text{ under } \mathbb{P}.
 \end{aligned}$$

- Any VAR(1), with same Σ , can be reached. We can also take:
 $\alpha_t = \alpha_{0,t} + \alpha_1 w_t$. We get the historical dynamics:

$$w_{t+1} = (A_0 - \Sigma \alpha_{0,t}) + (A_1 - \Sigma \alpha_1) w_t + \xi_{t+1}$$

- We can choose $\alpha_{0,t}$ such that: $A_0 - \Sigma\alpha_{0,t} = \begin{pmatrix} \mu_{1,t} \\ \mu_2 \\ \mu_3 \end{pmatrix}$
($\mu_{1,t}, \mu_2, \mu_3$ given).
- In particular, we have:

$$\begin{aligned} s_{t+1} &= \mu_{1,t} + (A_{11} - \Sigma_1\alpha_1)w_t + \xi_{1,t+1} \\ &= \mu_{1,t} + \bar{A}_{11}w_t + \xi_{1,t+1} \text{ (say),} \end{aligned}$$

with $\bar{A}_{11}$ not constrained, or alternatively:

$$s_{t+1} - \nu_{t+1} = \mu_1 + \bar{A}_{11,1}(s_t - \nu_t) + \bar{A}_{11,2}c_t + \bar{A}_{11,3}r_t + \xi_{1,t+1}$$

with

$$\mu_{1,t} := \mu_1 + \nu_{t+1} - \bar{A}_{11,1}\nu_t.$$

- In other words any historical seasonal function ν_t can be reached by choosing μ_{1t} , i.e. $\alpha_{0,t}$.

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- Model 3:

$$\begin{aligned}\mathbb{P} : \quad s_{t+1} &= a_{10} + s_t - c_t + r_t + \sigma_1 \varepsilon_{1,t+1} \\ c_{t+1} &= a_{20} + a_{21} c_t + \sigma_2 \varepsilon_{2,t+1} \\ r_{t+1} &= a_{30} + a_{31} r_t + \sigma_3 \varepsilon_{3,t+1}\end{aligned}$$

$$\begin{aligned}\mathbb{Q} : \quad s_{t+1} &= -\frac{\sigma_1^2}{2} + s_t - c_t + r_t + \sigma_1 \xi_{1,t+1} \\ c_{t+1} &= a_{20}^* + a_{21} c_t + \sigma_2 \xi_{2,t+1} \\ r_{t+1} &= a_{30} + a_{31} r_t + \sigma_3 \xi_{3,t+1},\end{aligned}$$

$\text{corr}(\varepsilon_{1t}, \varepsilon_{2t}) = \rho_1$, $\text{corr}(\varepsilon_{2t}, \varepsilon_{3t}) = \rho_2$, $\text{corr}(\varepsilon_{1t}, \varepsilon_{3t}) = \rho_3$ (idem for the ξ_{it} 's). We also have $\log F_{t,T}$ as affine functions of s_t, c_t, r_t .

- In these models s_t, c_t are latent, the parameters are estimated by the ML method. The likelihood functions are computed by the Kalman filter.
- The measurement equations are of the form:
 $\log F_{t,T} = \text{affine functions of the state variables} + \text{error}$ (for various values of t and T).
- State variables: s_t or (s_t, c_t) , (the $\mathbb{P}$ and $\mathbb{Q}$ dynamics of r_t is estimated separately).

Table XIII
Cross-Section Comparison Between Models 1, 2, and 3 Out of Sample
Copper Data 7/29/88 to 6/13/95

Model Contract	RMSE			Mean Error		
	1	2	3	1	2	3
Panel A: In Cents						
F2	4.617	1.311	1.27	1.75	-0.155	-0.142
F4	1.17	0.548	0.539	0.327	0.116	0.119
F6	1.133	0.616	0.621	-0.194	-0.105	-0.098
F8	1.824	0.534	0.482	0.257	-0.042	-0.064
All	2.612	0.819	0.794	0.535	-0.046	-0.046
Panel B: In Percentage						
F2	3.929	1.103	1.073	1.404	-0.151	-0.135
F4	1.063	0.483	0.474	0.284	0.093	0.094
F6	1.064	0.568	0.572	-0.188	-0.102	-0.097
F8	1.78	0.512	0.462	0.213	-0.036	-0.058
All	2.284	0.713	0.692	0.428	-0.049	-0.049

Out of sample refers to maturities not used at the estimation stage. Source: [Schwartz \(1997\)](#).

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A Gaussian VAR Model

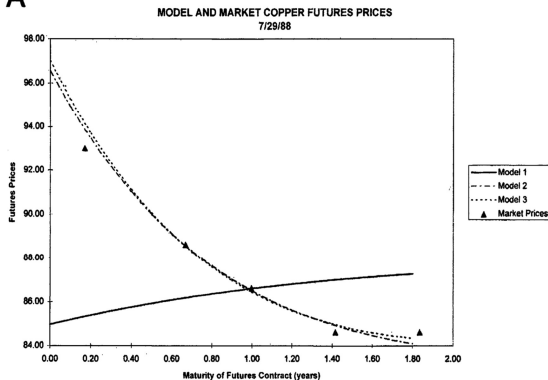
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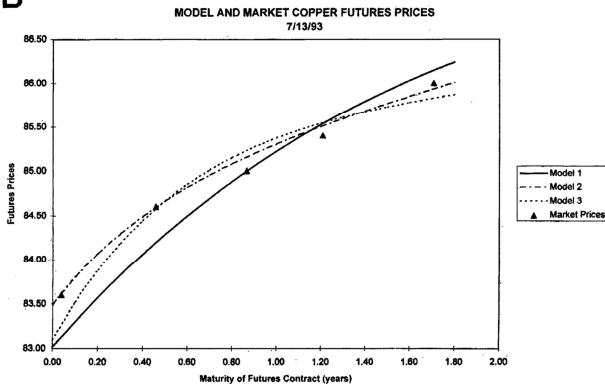
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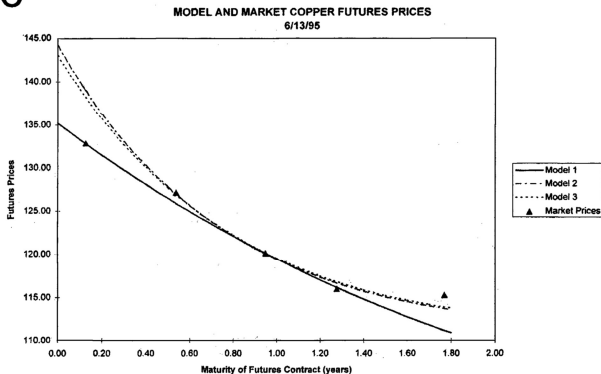
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Chapter II

MODELLING OF SPOT AND FORWARD ELECTRICITY PRICES

– Agenda –

- II.1 Specific features of electricity markets. (Slide 32)
- II.2 Modelling approach. (Slide 35)
- II.3 Pricing. (Slide 48)
- II.4 Application. (Slide 51)

II.1 – SPECIFIC FEATURES OF ELECTRICITY MARKETS

□ Spikes:

- we observe upward jumps shortly followed by a downward move (that can be captured by the introduction of switching regimes).

□ Electricity is not storable:

- no convenience yields, and no ICC for the pricing formula.

□ Seasonality:

- with daily observations, we have a daily seasonality (depending on the day in the week) and a monthly seasonality in the historical and R.N. world.

□ Continuous delivery:

- the price at date t of a Forward contract delivering a given number of kilowatt-hours during (some hours of) the days from $(t + H)$ to $(t + H + D)$
- can be derived from the price of a virtual forward contract providing one megawatt-hour on each day in $\{t + H, \dots, t + H + D\}$.

II.1 – SPECIFIC FEATURES OF ELECTRICITY MARKETS

□ Continuous delivery (cont.):

- The associated Forward (per-day) price $\tilde{\Phi}_t$ (say) at t is such that the value at t of the payoffs (i.e., the value of the contract):

$(S_{t+H} - \tilde{\Phi}_t)$ on day $(t + H)$,

$(S_{t+H+1} - \tilde{\Phi}_t)$ on day $(t + H + 1)$,

...

$(S_{t+H+D-1} - \tilde{\Phi}_t)$ on day $(t + H + D - 1)$,

is zero.

□ Notation:

- S_t is the date- t spot price of one megawatt/hour.
- $B(t, h)$ = price at t of a zero-coupon bond maturing at $t + h$.
- $F_{t,t+h}$ = Futures price at t (or forward price if the interest rates are exogenous) of one megawatt/hour delivered at $t + h$.

II.2 – MODELLING APPROACH

□ Notation:

$$\begin{aligned}s_t &:= (\log S_t - \nu_{1,t}) \text{ } (\nu_{1,t} \text{ catch seasonal effects}); \\ y_t &:= (Y_t - \nu_{2,t}) \text{ } (m\text{-dimensional (demand-like) factor}); \\ z_t &= \text{regime-indicator variable valued in } (e_1, \dots, e_J); \\ w_t &= (z_t', s_t, y_t')'; \\ w_t^e &= (\nu_{1,t}, \nu_{2,t}')' \text{ (exogenous factor)};\end{aligned}$$

□ Historical dynamics:

$$\begin{cases} s_{t+1} &= \mu' z_{t+1} + \varphi_1(s_t - \mu' z_t) + \varphi_2' y_t + u_{t+1} \\ y_{t+1} &= \psi_0 + \psi_1(s_t - \mu' z_t) + \psi_2' y_t + v_{t+1} \end{cases} \quad (\text{II.1})$$

$$\text{where } \begin{pmatrix} u_{t+1} \\ v_{t+1} \end{pmatrix} = \Sigma^{1/2}(z_{t+1}, z_t) \varepsilon_{t+1} \quad (\text{II.2})$$

- ε_{t+1} being a standard Gaussian white noise process of size $(m+1)$, and $\Sigma^{1/2}(z_{t+1}, z_t)$ a symmetric positive definite matrix function of (z_{t+1}, z_t) .
- We also assume that $\mathbb{P}(z_{t+1} = e_j \mid z_t, w_t)$ depends on z_t and s_t (but not on y_t). We denote $\pi_{i,j,t} = \mathbb{P}(z_{t+1} = e_j \mid z_t = e_i, s_t)$.
- Thus, (z_t) is non-homogeneous under $\mathbb{P}$ (not affine).

II.2 – MODELLING APPROACH

- Spikes: We denote $s_t^* = s_t - \mu'z_t$ and we have:

$$\begin{cases} s_t^* &= \varphi_1 s_{t-1}^* + \varphi_2' y_{t-1} + u_t \\ y_t &= \psi_0 + \psi_1 s_{t-1}^* + \psi_2' y_{t-1} + v_t. \end{cases}$$

- If z_t is in a spike state, i.e. $\mu'z_t$ "large", then $s_t = \mu'z_t + s_t^*$ is large;
- But, if the probability to stay in a spike between two subsequent dates is small, then $\mu'z_{t+1}$ is "likely small" and $s_{t+1} = \mu'z_{t+1} + s_{t+1}^*$ is likely to decrease immediately.

- This is not the case if the model was:

$$s_t = \mu'z_t + \varphi_1 s_{t-1} + \varphi_2' y_{t-1} + u_t.$$

- Indeed, a large value $\mu'z_t$ would impact s_t and therefore s_{t+1} through the AR term $\varphi_1 s_t$ (except if φ_1 is small), even if $\mu'z_{t+1}$ is small.

II.2 – MODELLING APPROACH

- We also have the possibility of replicating successive jumps if the probability of staying in a "spike" state i (say) increases in s_t .
- We can consider, for instance, the following specification for the non-homogeneous probability to stay in the "spike" state:

$$\frac{1}{1 + \exp(a_i + b_i s_t)}, \quad b_i < 0.$$

- Assuming $x_{t+1} = (s_{t+1}, y'_{t+1})'$, system (II.1)-(II.2) can be written

$$x_{t+1} = \mu(z_{t+1}, z_t) + \Phi x_t + \Sigma^{1/2}(z_{t+1}, z_t) \varepsilon_{t+1} \quad (\text{II.3})$$

with

$$\mu(z_{t+1}, z_t) = \begin{pmatrix} \mu' z_{t+1} - \varphi_1(\mu' z_t) \\ \psi_0 - \psi_1(\mu' z_t) \end{pmatrix}, \quad \Phi = \begin{pmatrix} \varphi_1 & \varphi'_2 \\ \psi_1 & \psi'_2 \end{pmatrix}.$$

Prop. II.1 (Historical dynamics)

The historical conditional Laplace transform of $w_{t+1} = (z'_{t+1}, x'_{t+1})'$ is:

$$\begin{aligned}\varphi_t(u, v) &= \mathbb{E}_t \exp(u' z_{t+1} + v' x_{t+1}) \\ &= \exp[v' \Phi x_t + (A_{1t}, \dots, A_{Jt}) z_t], \text{ where} \\ A_{it} &= \log \left\{ \sum_{j=1}^J \pi_{ijt} \exp[u_j + v' \mu(e_j, e_i) + \frac{1}{2} v' \Sigma(e_j, e_i) v] \right\}\end{aligned}$$

- It is important to note that φ_t is not in general exponential-affine in $w_t = (z'_t, x'_t)$ and, therefore, w_t is not an affine process.
- Indeed, A_{it} depends on w_t through π_{ijt} .
- However, w_t turns out to be an affine process, if π_{ijt} does not depend on s_t , i.e. if z_t is an homogeneous Markov chain.

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- **Proof:** The historical conditional Laplace transform of $w_{t+1} = (z'_{t+1}, x'_{t+1})'$ is:

$$\begin{aligned}\varphi_t(u, v) &= \mathbb{E}_t [\exp(u' z_{t+1} + v' x_{t+1})] \\ &= \mathbb{E}_t \exp \left[u' z_{t+1} + v' \mu(z_{t+1}, z_t) + v' \Phi x_t + v' \Sigma^{1/2} (z_{t+1}) \varepsilon_{t+1} \right].\end{aligned}$$

- Taking first the conditional expectation given $(\underline{w}_t, z_{t+1})$ we get:

$$\begin{aligned}\varphi_t(u, v) &= \exp(v' \Phi x_t) \mathbb{E}_t \exp \left[u' z_{t+1} + v' \mu(z_{t+1}, z_t) + \frac{1}{2} v' \Sigma(z_{t+1}, z_t) v \right] \\ &= \exp \left[v' \Phi x_t + (A_{1t}, \dots, A_{Jt}) z_t \right],\end{aligned}$$

$$\text{where } A_{it} = \log \left\{ \sum_{j=1}^J \pi_{ijt} \exp \left[u_j + v' \mu(e_j, e_i) + \frac{1}{2} v' \Sigma(e_j, e_i) v \right] \right\}. \quad \blacksquare$$

II.2 – MODELLING APPROACH

Prop. II.2 (Risk-neutral dynamics)

The risk-neutral conditional Laplace transform of w_{t+1} , given $\underline{w}_t$, is:

$$\varphi_t^{\mathbb{Q}}(u, v) = \exp \left[v' \Phi x_t + (\bar{A}_{1t}, \dots, \bar{A}_{Jt}) z_t \right], \quad (\text{II.6})$$

where:

$$\begin{aligned} \bar{A}_{it} = \log \left\{ \sum_{j=1}^J \pi_{ijt} \exp[v' \Sigma^{1/2}(e_j; e_i) \Gamma(e_j, e_i, x_t) \right. \\ \left. + \frac{1}{2} v' \Sigma(e_j, e_i) v + \delta_j(e_i, s_t) + u_j + v' \mu(e_j, e_i) \right\} \end{aligned}$$

and, therefore, the RN dynamics of (w_t) is (in general) not affine.

□ **Proof:** The RN conditional Laplace transform of w_{t+1} is :

$$\begin{aligned} \varphi_t^{\mathbb{Q}}(u, v) &= \mathbb{E}_t^{\mathbb{Q}} \exp \left(u' z_{t+1} + v' x_{t+1} \right) \\ &= \mathbb{E}_t \exp \left(\Gamma'_{t+1} \varepsilon_{t+1} - \frac{1}{2} \Gamma'_{t+1} \Gamma_{t+1} + \delta'_{t+1} z_{t+1} + u' z_{t+1} + v' x_{t+1} \right), \end{aligned}$$

with $\Gamma_{t+1} = \Gamma(z_{t+1}, z_t, x_t)$, $\delta_{t+1} = \delta(z_t, s_t)$ and $\sum_{j=1}^J \pi_{ijt} \exp[\delta_j(e_i, s_t)] = 1, \forall (e_i, s_t)$.

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□ **Proof (cont'd):** using the expression of x_{t+1} given by (II.3) we get:

$$\begin{aligned}
& \varphi_t^{\mathbb{Q}}(u, v) \\
= & \mathbb{E}_t \exp \left\{ \Gamma'_{t+1} \varepsilon_{t+1} - \frac{1}{2} \Gamma'_{t+1} \Gamma_{t+1} + \delta'_{t+1} z_{t+1} + u' z_{t+1} + v' [\mu(z_{t+1}, z_t) + \Phi_{X_t} \right. \\
& \left. + \Sigma^{1/2}(z_{t+1}, z_t) \varepsilon_{t+1}] \right\} \\
= & \exp(v' \Phi_{X_t}) \times \mathbb{E}_t \exp \left[-\frac{1}{2} \Gamma'_{t+1} \Gamma_{t+1} + \frac{1}{2} (\Gamma_{t+1} + \Sigma^{1/2} v)' (\Gamma_{t+1} + \Sigma^{1/2} v) \right. \\
& \left. + \delta'_{t+1} z_{t+1} + u' z_{t+1} + v' \mu_{t+1} \right] \\
= & \exp(v' \Phi_{X_t}) \mathbb{E}_t \exp \left[v' \Sigma^{1/2} \Gamma_{t+1} + \frac{1}{2} v' \Sigma v + \delta'_{t+1} z_{t+1} + u' z_{t+1} + v' \mu_{t+1} \right] \\
= & \exp \left[v' \Phi_{X_t} + (\bar{A}_{1t} \dots \bar{A}_{Jt}) z_t \right],
\end{aligned}$$

with:

$$\bar{A}_{it} = \log \left\{ \sum_{j=1}^J \pi_{ijt} \exp \left[v' \Sigma^{1/2} (e_j, e_i) \Gamma(e_j, e_i, x_t) + \frac{1}{2} v' \Sigma(e_j, e_i) v + \delta_j(e_i, s_t) + u_j + v' \mu(e_j, e_i) \right] \right\} \blacksquare$$

II.2 – MODELLING APPROACH

- Equivalently, we would like to have the system:

$$\begin{cases} \tilde{s}_{t+1} &= \mu^* z_{t+1} + \varphi_1^*(\tilde{s}_t - \mu^* z_t) + \varphi_2^* \tilde{y}_t + u_{t+1}^* \\ \tilde{y}_{t+1} &= \psi_0^* + \psi_1^*(\tilde{s}_t - \mu^* z_t) + \psi_2^* \tilde{y}_t + v_{t+1}^* \end{cases}$$

with $\begin{pmatrix} u_{t+1}^* \\ v_{t+1}^* \end{pmatrix} = \Sigma^{1/2}(z_{t+1}, z_t) \varepsilon_{t+1}^*$, and where ε_t^* is a standard

Gaussian white noise under $\mathbb{Q}$, and z_t is an homogeneous Markov chain under $\mathbb{Q}$, with transition probabilities π_{ij}^* .

- It turns out that from any $\mathbb{P}$ dynamics defined by (II.3) and π_{ijt} , we can reach any $\mathbb{Q}$ dynamics defined by (II.7) and π_{ij}^* , with a SDF of the form (II.4), provided that Γ_{t+1} and δ_{t+1} are well chosen.

Prop. II.3 (Affine risk-neutral dynamics)

If the historical dynamics is defined by system (II.3) and π_{ijt} , we obtain the RN dynamics defined by (II.7) and π_{ij}^ if we choose:*

$$\begin{aligned} \delta_j(e_i, s_t) &= \log \frac{\pi_{ij}^*}{\pi_{ijt}} \\ \Gamma(e_j, e_i, x_t) &= \Sigma^{-1/2}(e_j, e_i)[(\Phi^* - \Phi)x_t \\ &\quad + \Phi^*(\nu_t - \tilde{\nu}_t) + \mu^*(e_j, e_i) - \mu(e_j, e_i)]. \end{aligned} \tag{II.8}$$

II.2 – MODELLING APPROACH

- **Proof:** Given (II.7) and π_{ij}^* , the RN conditional Laplace transform of $\tilde{w}_{t+1} = (\tilde{x}_{t+1}, z_{t+1})$ is:

$$\begin{aligned} & \mathbb{E}_t^{\mathbb{Q}} \exp(u' z_{t+1} + v' \tilde{x}_{t+1}) \\ &= \exp(v' \Phi^* \tilde{x}_t) \mathbb{E}_t^{\mathbb{Q}} \exp \left[u' z_{t+1} + v' \mu^*(z_{t-1}, z_t) + v' \Sigma^{1/2}(z_{t+1}, z_t) \varepsilon_{t+1}^* \right] \\ &= \exp \left[v' \Phi^* \tilde{x}_t + (A_1^* \dots A_J^*) z_t \right], \end{aligned}$$

$$\text{with } A_i^* = \log \left\{ \sum_{j=1}^J \pi_{ij}^* \exp[u_j + v' \mu^*(e_j, e_i) + \frac{1}{2} v' \Sigma(e_j, e_i) v] \right\}.$$

- Therefore the RN conditional Laplace transform of w_{t+1} implied by (II.7) and π_{ij}^* is, using $\tilde{x}_t = x_t + \nu_t - \tilde{\nu}_t$, given by:

$$\exp[v' \Phi^* (\nu_t - \tilde{\nu}_t) + v' \Phi^* x_t + (A_1^* \dots A_J^*) z_t].$$

II.2 – MODELLING APPROACH

- **Proof (cont'd):** Comparing with $\varphi_t^{\mathbb{Q}}(u, v)$ given in Proposition II.2, we see that they are identical if:

$$i) \pi_{ijt} \exp[\delta_j(e_i, s_t)] = \pi_{ij}^*;$$

$$ii) \Sigma^{1/2}(e_j, e_i) \Gamma(e_j, e_i, x_t) = (\Phi^* - \Phi)x_t + \Phi^*(\nu_t - \tilde{\nu}_t) + \mu^*(e_j, e_i) - \mu(e_j, e_i).$$

- So, for a given $\mathbb{P}$ -dynamics characterized by (π_{ijt}, Φ, μ) and ν_t , and any
- wished (affine) RN dynamics characterized by $(\pi_{ij}^*, \Phi^*, \mu^*)$ and $\tilde{\nu}_t$,
- we can adjust the (market price parameters of the) SDF in the required way by choosing the δ_j satisfying *i)* and Γ satisfying *ii)*:

$$\delta_j(e_i, s_t) = \log \frac{\pi_{ij}^*}{\pi_{ijt}},$$

$$\Gamma(e_j, e_i, x_t) = \Sigma^{-1/2}(e_j, e_i) \times \\ [(\Phi^* - \Phi)x_t + \Phi^*(\nu_t - \tilde{\nu}_t) + \mu^*(e_j, e_i) - \mu(e_j, e_i)].$$

- Note that the δ_j thus defined automatically satisfy the constraints:

$$\sum_{j=1}^J \pi_{i,j,t} \exp[\delta_j(e_i, s_t)] = 1.$$

II.2 – MODELLING APPROACH

In this context, we can easily compute the RN conditional Laplace transform of $\tilde{w}_{t+1} = (\tilde{x}_{t+1}, z'_{t+1})'$.

Prop. II.4 (Affine Risk Neutral Laplace Transform)

Under the conditions of Proposition II.3, the RN conditional Laplace transform of $\tilde{w}_{t+1} = (\tilde{x}'_{t+1}, z'_{t+1})'$ is:

$$\begin{aligned}\tilde{\varphi}^Q(u, v) &= \mathbb{E}_t^Q \exp(u' z_{t+1} + v' \tilde{x}_{t+1}) \\ &= \exp[v' \Phi^* \tilde{x}_t + (A_1^* \dots A_J^*) z_t]\end{aligned}$$

$$\text{with : } A_i^* = \log \left\{ \sum_{j=1}^J \pi_{ij}^* \exp[u_j + v' \mu^*(e_j, e_i) + \frac{1}{2} v' \Sigma(e_j, e_i) v] \right\} \quad (\text{II.9})$$

So this conditional Laplace transform is of the form $\exp[a'(u, v) \tilde{w}_t + b(u, v)]$ with

$$a'(u, v) = (v' \Phi^*, A_1^* \dots, A_J^*) \text{ and } b(u, v) = 0.$$

and $\tilde{w}_{t+1}$ is an affine process.

II.3 – PRICING

We have $\tilde{\Phi}_t = \tilde{\Phi}_{t,t+H} = F_{t,t+H}$ ($D = 1$). Forward price at t of S_{t+H} :

$$\begin{aligned} F_{t,t+H} &= \mathbb{E}_t^{\mathbb{Q}}(S_{t+H}) = \mathbb{E}_t^{\mathbb{Q}}[\exp(\tilde{\nu}_{1,t+H} + \tilde{s}_{t+H})] \\ &= \exp(\tilde{\nu}_{1,t+H}) \mathbb{E}_t^{\mathbb{Q}}[\exp(\tilde{s}_{t+H})] \end{aligned}$$

$$\begin{aligned} \text{since } \tilde{w}_t &= (\tilde{x}'_t, z'_t) \text{ is affine} \\ \text{with } \tilde{x}'_t &= (\tilde{s}'_t, \tilde{y}'_t) \end{aligned}$$

and we can use the multi-horizon Laplace transform formulas.

Prop. II.5 (The Forward pricing formula)

The Forward price at time t , for a residual maturity h , $h = 1, \dots, H$, is

$$F_{t,t+h} = \exp(\tilde{\nu}_{1,t+h} + c'_h \tilde{w}_t)$$

where c_h is obtained recursively from

$$c_1 = a(e_1), \quad c_h = a(c_{h-1}), \quad h = 2, \dots, H.$$

and where $a(\cdot)$ is the function defined in Proposition II.4, or equivalently,

$$\log F_{t,t+h} = \tilde{\nu}_{1,t+h} + c_h^{(1)} \tilde{s}_t + c_h^{(2)'} \tilde{y}_t + c_h^{(3)'} z_t,$$

where c'_h has been partitioned into $c'_h = (c_h^{(1)}, c_h^{(2)'}, c_h^{(3)'})$.

II.3 – PRICING

Using $\log S_t = \nu_{1,t} + s_t = \tilde{\nu}_{1,t} + \tilde{s}_t$ and $Y_t = \nu_{2,t} + y_t = \tilde{\nu}_{2,t} + \tilde{y}_{2,t}$:

Corollary 1

$$\begin{aligned}\log F_{t,t+h} = & \tilde{\nu}_{1,t+h} + c_h^{(1)}(\nu_{1,t} - \tilde{\nu}_{1,t}) + c_h^{(2)'}(\nu_{2,t} - \tilde{\nu}_{2,t}) \\ & + c_h^{(1)}s_t + c_h^{(2)'}y_t + c_h^{(3)'}z_t\end{aligned}$$

- Therefore, the historical exogenous part of $\log F_{t,t+h}$, is:

$$m_{t,t+h} = \tilde{\nu}_{1,t+h} + c_h^{(1)}(\nu_{1,t} - \tilde{\nu}_{1,t}) + c_h^{(2)'}(\nu_{2,t} - \tilde{\nu}_{2,t}).$$

- In particular, the historical seasonally adjusted values of $F_{t,t+h}$ are:

$$f_{t,t+h} = \exp \left[\log F_{t,t+h} - m_{t,t+h} \right] = \exp(c_h^{(1)}s_t + c_h^{(2)'}y_t + c_h^{(3)'}z_t)$$

- If we assume that the seasonal patterns are identical in the historical and the RN world [see Lucia and Scharwtz (2000)], i.e. $\tilde{\nu}_{1,t} = \nu_{1,t}$, $\tilde{\nu}_{2,t} = \nu_{2,t}$, we get $m_{t,t+h} = \tilde{\nu}_{1,t+h}$.
- In other words, the seasonal part of $\log F_{t,t+h}$ is the same as the one of $\log S_{t+h}$, and, therefore it only depends on the delivery date $t+h$ and not on the present date t .

- We have also possibility to price European (Call, here below) options on spot prices :

$$\begin{aligned}C_S(t, K, h) &= B(t, h) \mathbb{E}_t^{\mathbb{Q}} (S_{t+h} - K)^+ \\&= B(t, h) \mathbb{E}_t^{\mathbb{Q}} \left[\exp \left(\tilde{\nu}_{1,t+h} + \tilde{s}_{t+h} \right) - K \right]^+, \end{aligned}$$

or on Forward prices :

$$\begin{aligned}C_F(t, K, h) &= B(t, h) \mathbb{E}_t^{\mathbb{Q}} [F(t+h, t+h+k) - K]^+ \\&= B(t, h) \mathbb{E}_t^{\mathbb{Q}} \left[\exp \left(\tilde{\nu}_{1,t+h+k} + c'_k \tilde{w}_{t+h} \right) - K \right]^+ . \end{aligned}$$

Lecture 2

- Electricity markets
- Modelling approach
- Pricing
- Application**

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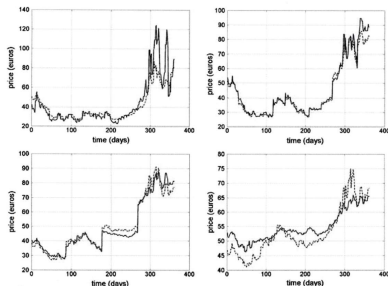


Fig.2 Result of forward reconstruction: realized forward price (solid line) versus reconstructed forward prices (dashed line) for the year 2007 of calibration

II.4 – APPLICATION - Model and estimation method

- Estimation : ν_t^s is estimated from two years data (January 1 2006, to December 31, 2007).
- The parameters of the historical dynamics are obtained by ML, the likelihood function being computed by the Kitagawa-Hamilton algorithm.
- The R.N. parameters obtained by minimizing :

$$J(\theta^*, \{y_t\}_{t=1, T}) = \sum_{t=1}^T \sum_{i=1}^{N_t} \|\log F_{t,t+H_{it},t+H_{it}+D_i}^o - \log F_{t,t+H_{it},t+H_{it}+D_i}\|^2$$

where N_t is the number of Forward contract observed at t , $F_{t,t+H_{it},t+H_{it}+D_i}^o$ denotes the observed ones, while $F_{t,t+H_{it},t+H_{it}+D_i}$ is the theoretical counterparts.

- the z_t^i 's are the smoothed values obtained from the estimation of the historical dynamics.

II.4 – APPLICATION - Results

Parameter	Init. value	Estimated value	95% Confidence interval
μ_1	-0.7033	0.0858	[0.0854; 0.0862]
μ_2	0.6385	-0.2393	[-0.2398; -0.2387]
μ_3	1.9803	0.2306	[0.2281; 0.2331]
φ_{11}	0	0.6541	[0.6537; 0.6546]
φ_{12}	0	0.0946	[0.0939; 0.0952]
φ_{13}	0	0.1433	[0.1429; 0.1437]
σ_1	0.1137	0.0127	[0.0127; 0.0127]
σ_2	0.1137	0.0318	[0.0316; 0.0320]
σ_3	0.1137	0.2317	[0.2288; 0.2346]

Table 1 Estimated historical parameters

(a) Probability transition

$$\begin{bmatrix} 0.9317 & 0.0563 & 0.0120 \\ 0.3215 & 0.6785 & 0 \\ 0 & 0.2073 & 0.7927 \end{bmatrix}$$

(b) Invariant probability

Regime	Probability
1	0.7873
2	0.1672
3	0.0456

Table 2 (a) Estimated historical transition probability and (b) resulting invariant probability

II.4 – APPLICATION - Results

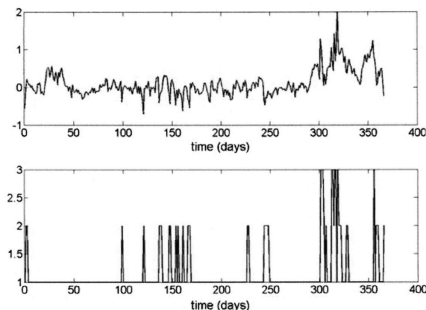


Fig.3 Observed s_t (up) and estimated state variable z_t (down)

Parameter	Init. value	Estimated value
μ_1^*	0.0858	-0.2232
μ_2^*	-0.2393	-0.1898
μ_3^*	0.2306	0.7087
φ_{11}^*	0.6541	0.9569
φ_{12}^*	0.0946	0.0442
φ_{13}^*	0.1433	-0.0028
φ_{21}^*	0	-0.0110
ψ_1^*	0	-0.0271
ψ_2^*	0	-0.0149

Table 3 Estimated risk-neutral parameters

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(a) Probability transition

0.9595	0.0356	0.0049
0.3335	0.6665	0.0000
0.0000	0.3488	0.6512

(b) Invariant probability

Regime	Probability
1	0.8807
2	0.1069
3	0.0124

Table 4 Estimated risk-neutral transition probability

Chapter III

SIMULTANEOUS MODELLING OF SEVERAL COMMODITY MARKETS

– Agenda –

III.1 A General Affine Framework. (Slide 61)

III.2 A Gaussian VAR Model. (Slide 63)

III.3 A Regime Switching Gaussian VAR Model. (Slide 66)

III.1 – A GENERAL AFFINE FRAMEWORK

Notation

- $w_t = (\log S_{1,t}, \dots, \log S_{N,t}, c_{1,t}, \dots, c_{N,t}, y'_t, r_t, z'_t)'$;
- $s_{n,t} = \log S_{n,t}$ $n = 1, \dots, N$;
- $S_{n,t}$: spot "cum convenience" price of commodity n at t ;
- $c_{n,t}$: convenience yield of commodity n ;
- $\tilde{S}_{n,t}$: "ex convenience" price, $S_{n,t} = \exp(c_{n,t})\tilde{S}_{n,t}$;
- r_t : risk-free interest rate between t and $t + 1$;
- y_t : factor of size K ;
- z_t : regime indicator variable, valued in $\{e_1, \dots, e_J\}$.

III.1 – A GENERAL AFFINE FRAMEWORK

Internal Consistency Conditions (ICC)

- Let us consider now the ICC for storable commodities:

$$\tilde{S}_{n,t} = \exp(-r_t) \mathbb{E}_t^{\mathbb{Q}} S_{n,t+1} \text{ or } \mathbb{E}_t^{\mathbb{Q}} S_{n,t+1} = \exp(r_t - c_{n,t}) S_{n,t}.$$

- Or, denoting $r_{n,t+1} = \log \frac{S_{n,t+1}}{S_{n,t}}$ the geometric return:

$$\mathbb{E}_t^{\mathbb{Q}} \exp(r_{n,t+1}) = \exp(r_t - c_{n,t}).$$

- Let us assume that $\{w_t\}$ is $\mathbb{Q}$ -affine, i.e.:

$$\mathbb{E}_t^{\mathbb{Q}} \exp(u' w_{t+1}) = \exp[a'(u) w_t + b(u)],$$

then $\mathbb{E}_t^{\mathbb{Q}} S_{n,t+1} = \mathbb{E}_t^{\mathbb{Q}} \exp(e_n' w_{t+1}) = \exp[a'(e_n) w_t + b(e_n)]$. Then, ICC:

$$\mathbb{E}_t^{\mathbb{Q}} S_{n,t+1} = \exp(r_t - c_{n,t} + s_{n,t}) = \exp[(e_{2N+K+1} - e_{N+n} + e_n)' w_t],$$

where e_{2N+K+1} is the selection vector of r_t . This implies $b(e_n) = 0$ and $a(e_n) = e_{2N+K+1} - e_{N+n} + e_n$. Specifying a SDF gives the $\mathbb{P}$ -dynamics (no constraint on the SDF).

III.2 – A GAUSSIAN VAR MODEL

□ $w_t = (s_{1,t}, \dots, s_{N,t}, c_{1,t}, \dots, c_{N,t}, r_t)$; no regimes or extra factors.

□ Under the $\mathbb{Q}$ -measure:

$$w_{t+1} = A_0 + A_1 w_t + \xi_{t+1}, \xi_t \stackrel{\mathbb{Q}}{\sim} IIN(0, \Sigma),$$

and the loadings of the associated Laplace transform are:

$$a(u) = A_1' u, \quad b(u) = A_0' u + \frac{1}{2} u' \Sigma u;$$

□ ICC (for any $n = 1, \dots, N$):

i) $a(e_n) = e_{2N+1} - e_{N+n} + e_n \Rightarrow$ the first N rows of A_1 are $e_{2N+1} - e_{N+n} + e_n$;

ii) $b(e_n) = 0 \Rightarrow$ the first N elements of A_0 are $-\frac{1}{2}\sigma_n^2$.

□ The ICC are satisfied iff the first N equations of the VAR are:

$$s_{n,t+1} = s_{n,t} - \frac{1}{2}\sigma_n^2 + r_t - c_{n,t} + \xi_{n,t+1}.$$

III.2 – A GAUSSIAN VAR MODEL

- In particular, denoting $r_{n,t+1} = s_{n,t+1} - s_{n,t}$, we have:

$$\mathbb{E}_t^{\mathbb{Q}} r_{n,t+1} = r_t - c_{n,t} - \frac{1}{2} \sigma_n^2.$$

and $r_{n,t+1} - r_t + c_{n,t} + \frac{1}{2} \sigma_n^2$ is a $\mathbb{Q}$ -martingale difference.

- The other equations of the model are free of restrictions. We might consider autonomous processes:

$$\begin{aligned} c_{t+1} &= C_0 + C_1 c_t + \eta_t, \\ r_{t+1} &= \alpha + \beta r_t + \varepsilon_t, \end{aligned}$$

(For electricity no ICC, no convenience yield, since not storable).

- Now, we can move back to the historical dynamics using the SDF:

$$\begin{aligned} \mathcal{M}_{t,t+1} &= \exp[-r_t + \alpha'_t w_{t+1} - \psi_t(\alpha_t)] \\ &= \exp[-r_t + \alpha'_t w_{t+1} - \alpha'_t A_0 - \alpha'_t A_1 w_t + \frac{1}{2} \alpha'_t \Sigma \alpha_t] \\ &= \exp[-r_t + \alpha'_t \xi_{t+1} + \frac{1}{2} \alpha'_t \Sigma \alpha_t], \end{aligned}$$

III.2 – A GAUSSIAN VAR MODEL

- We thus obtain the following conditional historical log-Laplace transform:

$$\begin{aligned}\psi_t(u) &= \psi_t^{\mathbb{Q}}(u - \alpha_t) - \psi_t^{\mathbb{Q}}(-\alpha_t) \\ &= u' A_1 w_t + u' A_0 + \frac{1}{2} (u - \alpha_t)' \Sigma (u - \alpha_t) - \frac{1}{2} \alpha_t' \Sigma \alpha_t \\ &= u' A_1 w_t + u' A_0 - u' \Sigma \alpha_t + \frac{1}{2} u' \Sigma u\end{aligned}$$

- If we assume an affine specification for the risk-correction term, namely $\alpha_t = \alpha_0 + \alpha_1 w_t$:

$$\begin{aligned}\psi_t(u) &= u' [A_0 - \Sigma \alpha_0 + (A_1 - \Sigma \alpha_1) w_t] + \frac{1}{2} u' \Sigma u, \\ &= u' (A_0 - \Sigma \alpha_0) + u' (A_1 - \Sigma \alpha_1) w_t + \frac{1}{2} u' \Sigma u,\end{aligned}$$

- and therefore we have:

$$\begin{aligned}w_{t+1} &= (A_0 - \Sigma \alpha_0) + (A_1 - \Sigma \alpha_1) w_t + \xi_t, \\ &= A_0^* + A_1^* w_t + \xi_t,\end{aligned}$$

where $A_0^* = (A_0 - \Sigma \alpha_0)$, $A_1^* = (A_1 - \Sigma \alpha_1)$, and with $\xi_t \stackrel{P}{\sim} N(0, \Sigma)$. We have a Gaussian VAR under the $\mathbb{P}$ measure, with different constant term and AR matrix but the same Σ .

III.3 – A REGIME SWITCHING GAUSSIAN VAR MODEL

□ $w_t = (s_{1,t}, \dots, s_{N,t}, c_{1,t}, \dots, c_{N,t}, y'_t, r_t, z'_t)'$, $w_t = (\tilde{w}'_t, z'_t)'$;

- y_t is of size K ;

□ Under the $\mathbb{Q}$ -measure:

- $\pi_{i,j} = \mathbb{Q}(z_{t+1} = e_j | z_t = e_i)$, $i = 1, \dots, J$ and $j = 1, \dots, J$;
- z_t is an autonomous and homogeneous Markov Chain;
- $\tilde{w}_t$ does not $\mathbb{Q}$ -cause z_t ;
- Conditional behavior of $\tilde{w}_{t+1}$ given z_{t+1} , $\underline{w}_t$:

$$\tilde{w}_{t+1} = L z_{t+1} + M z_t + \Phi \tilde{w}_t + \Omega(z_{t+1})\eta_{t+1},$$

with M , L matrices of size $(2N + K + 1) \times J$,

$\Omega(z_{t+1})\Omega(z_{t+1})' = \Sigma(z_{t+1})$, and $\eta_t \stackrel{\mathbb{Q}}{\sim} \text{IIN}(0, I)$.

□ Identification: we can assume, for instance, that the first column of M (or L) is zero. Notation: $\Sigma(e_j) = \Sigma_j$, l_j is the j^{th} column of L and m_i the i^{th} column of M .

III.3 – A REGIME SWITCHING GAUSSIAN VAR MODEL

Prop. III.1

The process $\{w_t\}$, with $w_t = (\tilde{w}_t', z_t')'$, is affine under the $\mathbb{Q}$ measure.

Proof:

□ If $z_t = e_j$:

$$\begin{aligned}\mathbb{E}_t^{\mathbb{Q}}[\exp(u' w_{t+1})] &= \mathbb{E}_t^{\mathbb{Q}}[\exp(u_1' \tilde{w}_{t+1} + u_2' z_{t+1})] \\ &= \sum_{j=1}^J \pi_{ij} \exp\left(u_1' l_j + u_1' m_i + u_1' \Phi \tilde{w}_t + \frac{1}{2} u_1' \Sigma_j u_1 + u_{2,j}\right)\end{aligned}$$

• and therefore:

$$\mathbb{E}_t^{\mathbb{Q}}[\exp(u' w_{t+1})] = \exp[u_1' \Phi \tilde{w}_t + \tilde{a}'(u) z_t]$$

$$\text{with } \tilde{a}'(u) = [\tilde{a}_1(u), \dots, \tilde{a}_J(u)]$$

$$\text{and } \tilde{a}_i(u) = u_1' m_i + \log \left[\sum_{j=1}^J \pi_{ij} \exp(u_1' l_j + \frac{1}{2} u_1' \Sigma_j u_1 + u_{2,j}) \right].$$

• In other words:

$$\mathbb{E}_t^{\mathbb{Q}}[\exp(u' w_{t+1})] = \exp[a'(u) w_t + b(u)]$$

$$\text{with } a'(u) = [u_1 \Phi, \tilde{a}'(u)], b(u) = 0. \blacksquare$$

III.3 – A REGIME SWITCHING GAUSSIAN VAR MODEL

□ Back to the historical dynamics, thanks to the SDF:

$$\begin{aligned}\mathcal{M}_{t,t+1} = & \exp[-r_t + \frac{1}{2}\alpha'(z_{t+1}, z_t, \tilde{w}_t)\alpha(z_{t+1}, z_t, \tilde{w}_t) \\ & + \alpha'(z_{t+1}, z_t, \tilde{w}_t)\eta_{t+1} + \delta'(z_t, \tilde{w}_t)z_{t+1}]\end{aligned}$$

with $\delta_j(e_i, \tilde{w}_t) = \log \frac{\pi_{i,j}}{\tilde{\pi}(e_i, e_j, \tilde{w}_t)}$. Then we have $\mathbb{E}_t^{\mathbb{Q}} M_{t,t+1}^{-1} = \exp(r_t)$, and the following holds:

Prop. III.2

If we take $\alpha(z_{t+1}, z_t, \tilde{w}_t) = \Omega^{-1}(z_{t+1})[\tilde{\Phi}\tilde{w}_t + \tilde{L}z_{t+1} + \tilde{M}z_t]$, we have under the $\mathbb{P}$ measure:

$$\begin{aligned}& (z_{t+1} | \underline{w}_t) \text{ driven by } \tilde{\pi}(e_i, e_j, \tilde{w}_t) \\ & \tilde{w}_{t+1} = (L - \tilde{L})z_{t+1} + (M - \tilde{M})z_t + (\Phi - \tilde{\Phi})\tilde{w}_t + \Omega(z_{t+1})\varepsilon_{t+1}, \\ & \varepsilon_{t+1} \stackrel{\mathbb{P}}{\sim} \text{IIN}(0, I),\end{aligned}$$

and where $L^* = L - \tilde{L}$, $M^* = M - \tilde{M}$, $\Phi^* = \Phi - \tilde{\Phi}$ are free. The conditional variance-covariance matrix $\Sigma(z_{t+1}) = \Omega(z_{t+1})\Omega'(z_{t+1})$ is the same as in the RN world.

III.3 – A REGIME SWITCHING GAUSSIAN VAR MODEL

Proof. Notations: $\Omega(z_{t+1}) = \Omega_{t+1}$, $\delta_t = \delta(z_t, \tilde{w}_t)$, $\alpha_{t+1} = \alpha(z_{t+1}, z_t, \tilde{w}_t)$.
Then, we have:

$$\begin{aligned}
 & \mathbb{E}_t^{\mathbb{P}} \exp(u'_1 \tilde{w}_{t+1} + u'_2 z_{t+1}) \\
 = & \mathbb{E}_t^{\mathbb{Q}} \exp \left[-\frac{1}{2} \alpha'_{t+1} \alpha_{t+1} - \alpha'_{t+1} \eta_{t+1} - \delta'(z_t, \tilde{w}_t) z_{t+1} + u'_1 \tilde{w}_{t+1} + u'_2 z_{t+1} \right] \\
 = & \exp(u'_1 M z_t + u'_1 \Phi \tilde{w}_t) \times \\
 & \mathbb{E}_t^{\mathbb{Q}} \exp \left[-\frac{1}{2} \alpha'_{t+1} \alpha_{t+1} - \alpha'_{t+1} \eta_{t+1} - \delta'_t z_{t+1} + u'_1 L z_{t+1} + u'_2 z_{t+1} + u' \Omega_{t+1} \eta_{t+1} \right] \\
 = & \exp(u'_1 M z_t + u'_1 \Phi \tilde{w}_t) \times \\
 & \mathbb{E}_t^{\mathbb{Q}} \exp \left[-\frac{1}{2} \alpha_{t+1} \alpha'_{t+1} + \frac{1}{2} (u'_1 \Omega_{t+1} - \alpha'_{t+1}) (u'_1 \Omega_{t+1} - \alpha'_{t+1})' + u'_1 L z_{t+1} + u'_2 z_{t+1} - \delta'_t z_{t+1} \right] \\
 = & \exp(u'_1 M z_t + u'_1 \Phi \tilde{w}_t) \times \\
 & \mathbb{E}_t^{\mathbb{Q}} \exp \left[-u'_1 \Omega_{t+1} \alpha_{t+1} + \frac{1}{2} u'_1 \Sigma_{t+1} u_1 + u'_1 L z_{t+1} + u'_2 z_{t+1} - \delta'_t z_{t+1} \right] \\
 = & \exp(u'_1 M z_t + u'_1 \Phi \tilde{w}_t - u'_1 \tilde{\Phi} \tilde{w}_t - u'_1 \tilde{M} z_t) \times \\
 & \mathbb{E}_t^{\mathbb{Q}} \exp \left[u'_1 (L - \tilde{L}) z_{t+1} + u'_2 z_{t+1} - \delta'_t z_{t+1} + \frac{1}{2} u'_1 \Sigma_{t+1} u_1 \right].
 \end{aligned}$$

III.3 – A REGIME SWITCHING GAUSSIAN VAR MODEL

Proof (cont'd). If $z_t = e_i$:

$$\begin{aligned} &= \exp \left[u'_1(m_i - \tilde{m}_i) + u'_1(\Phi - \tilde{\Phi})\tilde{w}_t \right] \times \\ &\quad \sum_{j=1}^J \pi_{i,j} \exp \left[\log \left(\frac{\tilde{\pi}(e_i, e_j, \tilde{w}_t)}{\pi_{i,j}} \right) + u'_1(l_j - \tilde{l}_j) + u_{2,j} + \frac{1}{2} u'_1 \Sigma_j u_1 \right] \\ &= \sum_{j=1}^J \tilde{\pi}(e_i, e_j, \tilde{w}_t) \exp \left[u'_1(l_j - \tilde{l}_j) + u'_1(m_i - \tilde{m}_i) + u'_1(\Phi - \tilde{\Phi})\tilde{w}_t + \frac{1}{2} u'_1 \Sigma_j u_1 + u_{2,j} \right], \end{aligned}$$

and comparing with the expression in the proof of Proposition III.1 gives the result. ■

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Chapter IV

BASIC CONCEPTS IN FIXED INCOME ANALYSIS

– Agenda –

- IV.1 Discount Factors. (Slide 73)
- IV.2 Interest Rates. (Slide 81)
- IV.3 The Term Structure of Interest Rates. (Slide 90)
- IV.4 From Zero-Coupon Bonds to Coupon Bonds. (Slide 95)
- IV.5 From Coupon Bonds to Zero-Coupon Bonds. (Slide 101)
- IV.6 Yield to Maturity and Return. (Slide 104)
- IV.7 Forward Rates and Forward Discount Factors. (Slide 111)
- IV.8 Extracting Yield Curves from Bond Prices. (Slide 121)

IV.1 – DISCOUNT FACTORS

- Receiving a dollar today is not the same as receiving it in a month or in a year:
 - ▶ indeed, money today can be put in a safe place (a bank) during any preferred/required period.
- That is, money in hands gives its holder the *option* to use it however she or he desires, including transferring it to the future through a deposit or an investment.
 - ▶ This option has value on its own.
- If we agree that households and investors value 1\$ today *more* than 1\$ in the future the question is then how much 1\$ in the future is worth in today's money.
 - ▶ The value of what 1\$ in the future would be in today's money is called *Discount Factor*.

IV.1 – DISCOUNT FACTORS

- Let us look at a concrete example: the U.S. government needs to borrow money from investors to finance its expenses. The government issue a number of securities, such as Treasury bills, notes, and bonds, to investors, receiving money today in exchange for money in the future.
- The U.S. government is extremely unlikely to default on its obligations, and thus the relation between purchase price and payoff of U.S. Treasury securities
 - ▶ reveals the market time value of money, that is,
 - ▶ the exchange rate between money today and money in the future.
- The following example illustrate this point.

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Discount Factors

Interest Rates

The Term Structure of
Interest Rates

From zero-coupon bonds to
coupon bonds

From coupon bonds to
zero-coupon bonds

Yield to Maturity and
Return

Forward Rates and Forward
Discount Factors

Extracting Yields
from Bond Prices

IV.1 – DISCOUNT FACTORS

□ Example 1.1:

- ▶ On August 10, 2006 the Treasury issued 182-day Treasury bills. The issuance market price was \$97.477 for \$100 of face value.
- ▶ That is, on August 10, 2006, investors were willing to buy for \$97.477 a government security that would pay \$100 on February 8, 2007.
- ▶ This Treasury bill would not make any other payment between the two dates (it is a zero-coupon bond).
- ▶ Thus, the ratio between purchase price and the payoff:

$$0.97477 = \frac{\$97.477}{\$100}$$

can be considered the market-wide discount factor between the two dates August 10, 2006 and February 8, 2007.

- ▶ That is, market participants were willing to exchange 0.97477 dollars on the first date for 1 dollar six months later.

IV.1 – DISCOUNT FACTORS

- **Definition :** *The discount factor between two dates, t and T , provides the term of exchange between a given amount of money at t versus a (certain) unit of money at a later date T . We denote the discount factor between these two date by $Z(t, T)$.*
 - ▶ In the above example, the two dates are $t = \text{August 10, 2006}$ and
 - ▶ $T = \text{February 8, 2007}$.
 - ▶ The discount factor is $Z(t, T) = 0.97477$
- In short, the discount factor $Z(t, T)$ records the time value of money between t and T .
- Since it is a value (namely, the value today of \$1 in the future), it is essentially a *price*: how much money somebody is willing to pay today in order to have \$1 in the future.

IV.1 – DISCOUNT FACTORS

- As we have seen, $Z(t, T)$ records the time value of money between t and T . At any given time t , the discount factor is lower, the longer the maturity T . That is given two dates T_1 and T_2 , with $T_1 < T_2$, it is always the case that:

$$Z(t, T_1) \geq Z(t, T_2), \quad (Z(t, t) = 1);$$

it is always the case that market participants prefer a \$1 sooner than later.

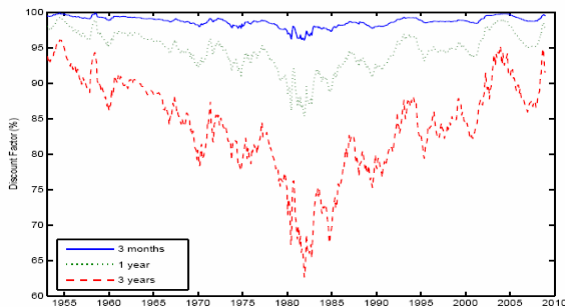
- **Example 1.2:** on August 10, 2006 the U.S. government also issued 91-day bills with a maturity date of November 9, 2006. The price was \$98.739 for \$100 of face value.
- Thus, denoting again $t = \text{August 10, 2006}$, now $T_1 = \text{November 9, 2006}$, and $T_2 = \text{February 8, 2007}$, we find that the discount factor $Z(t, T_1) = 0.98739$, which is higher than $Z(t, T_2) = 0.97477$.

IV.1 – DISCOUNT FACTORS

- The time series of $Z(t, T)$, for a constant time-to-maturity $T - t$, is time-varying.
- One of the variables that determines the variation over time (date t) of this value is inflation:
 - ▶ Higher expected inflation, makes less appealing money in the future so discounts go down.
- Inflation is not the only variable that explains discount factors: macro economic conditions, actions of the Central Bank as well as time-varying risk-aversion of investors may affect the relative supply and demand of bonds and thus their prices.

IV.1 – DISCOUNT FACTORS

Figure 2.1 Discount Factors



Source: Center for Research in Security Prices (CRSP)

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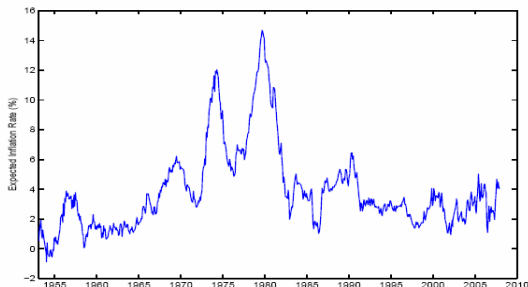
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Figure 1: Discount Factors. From Veronesi (2010, Chapter 2).

IV.1 – DISCOUNT FACTORS

Figure 2.2 Expected Inflation



Data Source: Bureau of Labor Statistics.

Figure 2: Expected Inflation. From Veronesi (2010, Chapter 2).

IV.2 – INTEREST RATES

- Interest rates are closely related to discount factors and are more similar to the concept of return on an investment:
 - ▶ yet it is more complicated, because it depends on the compounding frequency.
- The **compounding frequency of interest accruals** refers to the number of times within a year in which interests are paid on the invested capital.
- For a given interest rate, a higher compounding frequency results in a higher payoff.
- For a given payoff, a higher compounding frequency results in a lower interest rate.

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IV.2 – INTEREST RATES

- **Periodically Compounded Interest Rate** : is the concept of adding accumulated interest back to the principal.

Example 1.3: an amount of $P = 1500$ dollars (the principal) is deposited at date t in a bank paying an annual interest rate of 4.3%, compounded quarterly. Which is the value of the principal after 6 years ?

$$A_1 = 1500 \times (1 + 0.043/4), \text{ value of the principal after 1 quarter}$$

$$A_2 = A_1 \times (1 + 0.043/4) = 1500 \times (1 + 0.043/4)^2, \dots \text{ after 2 quarters}$$

$\vdots$

$$A = 1500 \times (1 + 0.043/4)^{4 \times 6} = 1938.8, \dots \text{ after 6 years.}$$

Remark : 4.3% is the **nominal rate** (unadjusted for compounding!), while the equivalent **effective annual rate** is 4.37% (adjusted for periodic compounding):

$$1500 \times (1 + 0.0437)^6 = 1938.8.$$

IV.2 – INTEREST RATES

□ Example 1.4 (*Semi-annual Compounding*):

- ▶ Let t = August 10, 2006, and let T = August 10, 2007 (one year later). Consider a year investment of $P = \$100$ at t with semi-annually compounded (annual) interest $r = 5\%$.
- ▶ This terminology means that after six months the investment grows to $\$102.5 = \$100 \times (1 + 5\%/2)$, which is then reinvested at the same rate for another 6 months, yielding at T :

$$\text{Payoff at } T: \$105.0625 = (\$100) \times (1 + r/2) \times (1 + r/2) = (\$100) \times (1 + r/2)^2$$

- ▶ Given that the initial investment is \$100, there are no cash flows to the investor during the period, and the payoff at T is risk free, the relation between money at t (\$100) and money at T ($= \$105.0625 = \text{payoff at } T$) establishes a discount factor between the two dates:

$$\frac{\$100}{\text{Payoff at } T} = \frac{1}{(1 + r/2)^2}$$

IV.2 – INTEREST RATES

- Let $r_2(t, T)$ denote the (annualized) semi-annually compounded interest rate between t and T . Then $r_2(t, T)$ defines a discount factor as:

$$Z(t, T) = \frac{1}{\left(1 + \frac{r_2(t, T)}{2}\right)^{2 \times (T-t)}}.$$

- The logic of this fact lies in the example above. The semi-annually compounded interest rate $r_2(t, T)$ defines a payoff at maturity T given by:

$$\text{Payoff at } T = \text{Investment at } t \times \left(1 + \frac{r_2(t, T)}{2}\right)^{2 \times (T-t)}$$

- Since the payoff at T is known at t , the relation between investment today at t and the payoff at T defines the time value of money, and $Z(t, T)$ given above defines the rate of exchange between money at T and money at t .
- Similarly, given a discount factor $Z(t, T)$, we can obtain the semiannually compounded interest rate. The following example illustrates this point.

□ **Example 1.5:**

- ▶ On March 1, 2001 (time t) the Treasury issued a 52-week Treasury bill, with maturity date $T = \text{February 28, 2002}$. The price of the Treasury bill was \$95.713.
- ▶ As we have learned, this price defines a discount factor between the two dates of $Z(t, T) = 0.95713$.
- ▶ At the same time, it also defines a semi-annually compounded interest rate equal to $r_2(t, T) = 4.43\%$. In fact, $\$95.713 \times (1 + 4.43\%/2)^2 = \100 .
- ▶ The semi-annually compounded interest rate can be computed from $Z(t, T) = 0.95713$ by solving for $r_2(t, T)$:

$$r_2(t, T) = 2 \times \left(\frac{1}{Z(t, T)^{\frac{1}{2}}} - 1 \right) = 2 \times \left(\frac{1}{0.95713^{\frac{1}{2}}} - 1 \right) = 4.43\% .$$

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IV.2 – INTEREST RATES

- Let $Z(t, T)$ be the discount factor between t and T . Then the semi-annually compounded interest rate $r_2(t, T)$ can be computed from the formula:

$$r_2(t, T) = 2 \times \left(\frac{1}{Z(t, T)^{\frac{1}{2 \times (T-t)}}} - 1 \right).$$

- More Frequent Compounding:

- Let the discount factor $Z(t, T)$ be given, and let $r_n(t, T)$ denote the (annualized) n -times compounded interest rate. Then $r_n(t, T)$ is defined by the equation:

$$Z(t, T) = \frac{1}{\left(1 + \frac{r_n(t, T)}{n} \right)^{n \times (T-t)}}.$$

Solving for $r_n(t, T)$, we obtain:

$$r_n(t, T) = n \times \left(\frac{1}{Z(t, T)^{\frac{1}{n \times (T-t)}}} - 1 \right).$$

IV.2 – INTEREST RATES

- **In summary** - The value A of the principal P after $m = T - t$ years at the nominal annual rate r compounded n times per year is:

$$A = P \times \left(1 + \frac{r}{n}\right)^{n \times m}.$$

- The associated **effective annual rate** is the rate r^{eff} such that:

$$(1 + r^{eff}) = \left(1 + \frac{r}{n}\right)^n \Rightarrow r^{eff} = \left(1 + \frac{r}{n}\right)^n - 1.$$

- **Present and Future Values** : Let us consider a nominal interest rate r per year, compounded n times per year. The **present value** $V_P(n, m)$ of 1 unit of money to be received in m years is:

$$V_P(n, m) = \left(1 + \frac{r}{n}\right)^{-n \times m}, \text{ over } m = T - t \text{ years}$$

The **future value** of 1 unit of money capitalized n times per year at the rate r for m years is:

$$V_F(n, m) = \left(1 + \frac{r}{n}\right)^{n \times m}, \text{ over } m = T - t \text{ year}; \quad V_F(n, m) = 1/V_P(n, m)$$

IV.2 – INTEREST RATES

- **Continuously Compounded Interest Rate** : if we consider a compounding period infinitely small, that is if $n \rightarrow \infty$, we have:

$$V_P(\infty, m) = \lim_{n \rightarrow \infty} V_P(n, m) = \exp(-r_\infty m), \text{ over } m \text{ years}$$

$$V_F(\infty, m) = \lim_{n \rightarrow \infty} V_F(n, m) = \exp(r_\infty m),$$

and r_∞ denotes the **nominal annual continuously compounded rate**. It is also called **force of interest**. The associated **effective annual rate** is:

$$r_\infty^{\text{eff}} = \exp(r_\infty) - 1.$$

- *The continuously compounded interest rate $r(t, T)$, obtained from $r_n(t, T)$ for n that increases to infinity, is given by the formula:*

$$Z(t, T) = \exp(-r(t, T)(T - t)).$$

Solving for $r(t, T)$, we obtain:

$$r(t, T) = -\frac{\ln(Z(t, T))}{T - t}.$$

IV.2 – INTEREST RATES

- **Simple Interest Rate (linear capitalization):** in contrast to periodically compounded interest rates, when we discount or capitalize using simple interest rates, the interest generated at each period is not added to the principal. We also refer to *simply compounded* rates.
- The future value of 1 unit of money invested for m years at the annual rate R is:

$$V_F^m = (1 + R \times m), \text{ over } m \text{ years, and } V_F^m = 1/V_P^m.$$

Example 1.6: Given at date t a principal of $P = 2500$ dollars and a simple annual interest rate $R = 12.99\%$, its future value after 3 months is:

$$V_F^{0.25} = 2500 \times (1 + 0.1299 \times 0.25) = 2580.6.$$

- Let us consider $n = 1$ (and $R > 0$), then:

$$V_F(1, m) = (1 + R)^m \geq V_F^m = (1 + R \times m).$$

The equality is attained when $m = 1$.

Example 1.7: Let us take $P = 2500$, $m = 5$, $n = 1$ and an annual rate $r = 0.03$. Depending on the kind of compounding (simply or periodically) we have $V_F^5 = 2875$ and $V_F(1, 5) = 2898.2$.

If $n = 4$, we have $V_F(4, 5) = 2903$ and, if $n = \infty$, $V_F(\infty, 5) = 2905$.

IV.3 – THE TERM STRUCTURE OF INTEREST RATES

- The **term structure of interest rates**, or **spot curve**, or **yield curve**, at a certain time t defines the relation between the level of interest rates and their time to maturity $T - t$.
- The **discount curve** at a certain time t defines instead the relation between the date- t discount factor $Z(t, T)$ and their time to maturity $T - t$.
- The term spread is the difference between long term interest rates (e.g. 10 year rate) and the short term interest rates (e.g. 3 month interest rate). The term spread depends on many variables: expected future inflation, expected growth of the economy, agents attitude towards risk, etc.
- The term structure varies over time, and may take different shapes.

IV.3 – THE TERM STRUCTURE OF INTEREST RATES

□ Example 1.8:

- ▶ On June 5, 2008, the Treasury issued 13-week, 26-week and 52-week bills at prices \$99.5399, \$99.0142, and \$97.8716, respectively.
- ▶ Denoting $t = \text{June 5, 2008}$, and T_1 , T_2 , and T_3 the three maturity dates, the implied discount factors are $Z(t, T_1) = 0.995399$, $Z(t, T_2) = 0.990142$, and $Z(t, T_3) = 0.978716$.
- ▶ The discount factor of longer maturities is lower than the one of shorter maturities. The question is then: how much lower is $Z(t, T_3)$, say, compared to $Z(t, T_2)$ or $Z(t, T_1)$?
- ▶ Translating the discount factors into annualized interest rates provides a better sense of the relative value of money across maturities. In this case, the continuously compounded interest rates are:

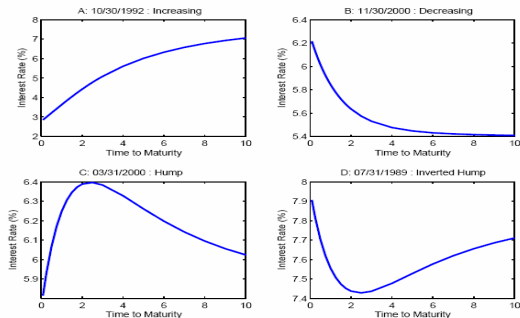
$$r(t, T_1) = -\frac{\ln(0.995399)}{0.25} = 1.8444\% ;$$

$$r(t, T_2) = -\frac{\ln(0.990142)}{0.5} = 1.9814\% ;$$

$$r(t, T_3) = -\frac{\ln(0.978716)}{1} = 2.1514\% .$$

IV.3 – THE TERM STRUCTURE OF INTEREST RATES

Figure 2.3 The Shapes of the Term Structure



Data Source: Center for Research in Security Prices

Figure 3: The Shapes of the Term Structure. From Veronesi (2010, Chapter 2).

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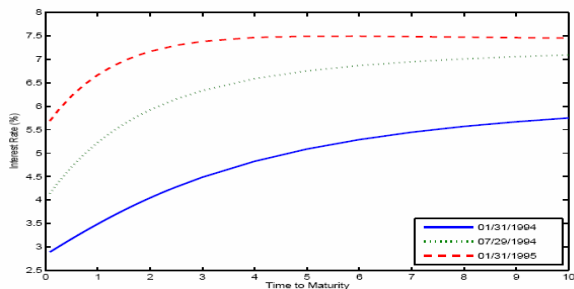
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IV.3 – THE TERM STRUCTURE OF INTEREST RATES

Figure 2.4 The Term Structure of Interest Rates on Three dates

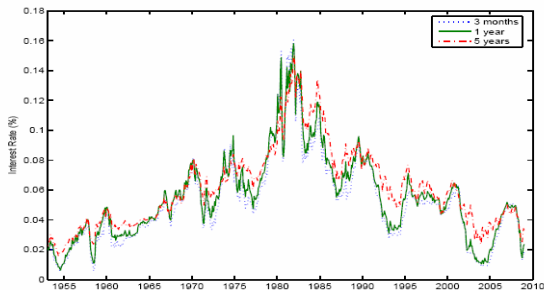


Data Source: Center for Research in Security Prices

Figure 4: The Term Structure on three dates. From Veronesi (2010, Chapter 2).

IV.3 – THE TERM STRUCTURE OF INTEREST RATES

Figure 2.5 The Term Structure over Time



Source: Center for Research in Security Prices

Figure 5: The Term Structure over time. From Veronesi (2010, Chapter 2).

IV.4 – FROM ZERO-COUPON BONDS TO COUPON BONDS

- A **zero-coupon bond** (ZCB) is the simplest possible bond :
 - ▶ it promises today (date t) a **single and predetermined** payment at the maturity date T ;
 - ▶ this payment (denoted C_T) is called **face value, par value** or **principal**. It is the amount of money paid to the bond holder at the maturity date.
- Let us denote by $P_z(t, T)$ the price at the date t (date of potential issue) of a zero-coupon bond, with face value $C_T = 100$, maturing at date T (the maturity date). Then we have:

$$P_z(t, T) = C_T \times Z(t, T) = 100 \times Z(t, T).$$

This means that from observed prices for zero coupon bonds we can compute the discount factors.

IV.4 – FROM ZERO-COUPON BONDS TO COUPON BONDS

- A **coupon bond** is a bonds which promises, not only the face value at the maturity date, but also other regular payments (coupons) between the date of issue and the maturity date (included).
- A coupon bond has multiple payment dates T_i , $\{1, \dots, n\}$, $T_1 < \dots < T_n$ (with $T_n = T$ the maturity date). The payment (coupon) at date T_i is given (in general) by the coupon rate c_i times the face value C_T . At T , the holder receives also C_T .
- In general, payments occur at regular intervals (6 months for T-bills, T-bonds) so that, for all i , $T_{i+1} - T_i = \Delta_i = \Delta$ for some fixed time interval Δ . If we measure time in "years", typical bonds have $\Delta \in \{0.25, 0.5, 1\}$ corresponding to quarterly, semi-annual or annual payments.

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IV.4 – FROM ZERO-COUPON BONDS TO COUPON BONDS

- In many cases coupon rates c_i are quoted as an annual rate even if they are paid more frequently $\Rightarrow$ the periodic coupon rate is $c_i \Delta$.
- Let us consider the (classical) so-called *bullet bonds* or *straight-coupon bonds* : here, all payments before the maturity date (T_i , $i < n$) are given by the product of the coupon rate (c_i) and the face value (C_T). The payment at the maturity date $T_n = T$ is the sum of the coupon plus the face value.
- More formally, given an annual coupon rate of c_i , the payments C_i (say) at the dates $T_1 < \dots < T_n$, per unit of face value (C_T), are:

$$C_i = \begin{cases} c_i \times C_T \times \Delta, & \forall i \in \{1, \dots, n-1\} \\ c_i \times C_T \times \Delta + C_T, & i = n. \end{cases}$$

For $c_i = 0$, $\forall i \in \{1, \dots, n\}$, we are back to a ZCB.

IV.4 – FROM ZERO-COUPON BONDS TO COUPON BONDS

- A coupon bond can be seen as a portfolio of ZCBs :
- ▶ $(c_1 \times \Delta)$ units of ZCBs maturing at T_1 , $(c_2 \times \Delta)$ units of ZCBs maturing at T_2 , etc.
- ▶ Indeed:

$$\begin{aligned}P_c(t, T) &= \sum_{i=1}^n C_i \times Z(t, T_i) \\&= \sum_{i=1}^n (c_i \times C_T \times \Delta) Z(t, T_i) + C_T Z(t, T_n) \\&= \sum_{i=1}^n (c_i \times \Delta) P_z(t, T_i) + P_z(t, T_n), \text{ otherwise A.O.!}\end{aligned}$$

(IV.1)

where we implicitly assume that all these ZCBs are traded in the market.

IV.4 – FROM ZERO-COUPON BONDS TO COUPON BONDS

- **Example 1.9:** Let us consider at date t , a bullet bond with $C_T = 100$, an annual coupon rate of 7% and time-to-maturity 3 years ($T - t = 3$ years).
- Suppose that, at date t , ZCBs with face value equal to 1 dollar (i.e., discount factors), are traded with residual maturity of 1, 2 and 3 years.
- Assume that the market prices are:

$$P_z(t, t+1y) = 0.98, \quad P_z(t, t+2y) = 0.94, \quad P_z(t, t+3y) = 0.90.$$

- The price of the bullet bond must be:

$$\begin{aligned} P_c(t, t+3y) &= (0.07 \cdot 100) \cdot 0.98 + (0.07 \cdot 100) \cdot 0.94 \\ &\quad + [(0.07 \cdot 100) + 100] \cdot 0.90 = 109.74. \end{aligned}$$

IV.4 – FROM ZERO-COUPON BONDS TO COUPON BONDS

- ▶ If $P_c(t, t + 3y) < 109.74 \Rightarrow$ the market provide an **arbitrage opportunity**, that is a **risk-free profit** : at date t sell 7 one-year, 7 two-year and 107 three-year ZCBs (earning 109.74), and buy the bullet bond at the market price $P_c(t, t + 3y)$.
- You obtain (for sure!) a positive net profit at t of $109.74 - P_c(t, t + 3y)$.
- ▶ If $P_c(t, t + 3y) > 109.74 \Rightarrow$ you have again an arbitrage opportunity! Why ?
- ▶ If at date t the market does not trade all relevant ZCBs that we need to replicate the coupon bond, we cannot justify relation (IV.1) via the replication argument.
- ▶ Nevertheless, the relation (based on the no-arbitrage principle) is still correct! If an investor has determined a discount function $Z(t, T_i)$ (using bond market information), then he/she can use it to implement (IV.1).

IV.5 – FROM COUPON BONDS TO ZERO-COUPON BONDS

- We can also go the other way around: If we have enough coupon bonds, we can compute the implicit value of zero coupon bonds from the prices of coupon bonds.
- Relation (IV.1) be used to estimate the discount factors $Z(t, T)$ for every maturity. The following example illustrates the reasoning:
- **Example I.10:** On $t = \text{June 30, 2005}$, the 6-month Treasury bill, expiring on $T_1 = \text{December 29, 2005}$, was trading at \$98.3607. On the same date, the 1 year to maturity, 2.75% Treasury note (semi-annual payments), was trading at \$99.2343. The maturity of the latter Treasury note is $T_2 = \text{June 30, 2006}$. Using (IV.1) we can write:

$$P_{bill}(t, T_1) = \$98.3607 = \$100 \times Z(t, T_1)$$

$$P_{bill}(t, T_2) = \$99.2343 = \$1.375 \times Z(t, T_1) + \$101.375 \times Z(t, T_2)$$

IV.5 – FROM COUPON BONDS TO ZERO-COUPON BONDS

- **Example (cont.):** We have two equations in two unknowns [the discount factors $Z(t, T_1)$ and $Z(t, T_2)$]:

from the first equation we obtain the discount factor $Z(t, T_1) = \frac{\$98.3607}{\$100} = 0.983607$; we can substitute this value into the second equation, and solve for $Z(t, T_2)$ to obtain $Z(t, T_2) = 0.965542$.

- The prices of coupon bonds, then, implicitly contain the information about the market time value of money. This procedure can be iterated forward to obtain additional terms.
- More precisely, if we have a sufficient amount of data, we can proceed in this fashion for every maturity, and obtain all of the discount factors $Z(t, T)$. This methodology is called the **bootstrap methodology**.

IV.5 – FROM COUPON BONDS TO ZERO-COUPON BONDS

- **Definition:** Let t be a given date. Let there be n coupon bonds, with coupon rates c_i , maturities T_i and coupon bond prices denoted by $P_c(t, T_i)$. Assume that maturities are at regular intervals of six months, that is, $T_1 = t + 0.5$ and $T_i = T_{i-1} + 0.5$. Then, the **bootstrap methodology** to estimate discount factors $Z(t, T_i)$ for every $i \in \{1, \dots, n\}$ is as follows:

1. The first discount factor $Z(t, T_1)$ is given by:

$$Z(t, T_1) = \frac{P_c(t, T_1)}{100 \times (1 + c_1/2)}.$$

2. Any other discount factor $Z(t, T_i)$ for $i \in \{2, \dots, n\}$ is given by:

$$Z(t, T_i) = \frac{P_c(t, T_i) - (c_i/2) \times 100 \left(\sum_{j=1}^{i-1} Z(t, T_j) \right)}{100 \times (1 + c_i/2)}.$$

IV.6 – YIELD TO MATURITY AND RETURN

- How can we measure the expected return on an investment in Treasury securities?
- Assuming the investor will hold the bond until maturity, computing the expected return on an investment in a bond is (more or less) simple, as the final payoff is known. In this case we talk about **yield to maturity** or **internal rate of return**.
- **Yield of a coupon bond** - Given at date t a coupon bond with maturity date T , price $P_c(t, T)$ and payments $\{C_1, \dots, C_n\}$ at dates $\{T_1, \dots, T_n\}$, respectively, the associated **annually compounded yield to maturity** is the value $Y_c(t, T)$ satisfying the equation:

$$P_c(t, T) = \sum_{i=1}^n C_i(1 + Y_c(t, T))^{-(T_i - t)}.$$

Note that the same discount rate is applied to all payments!

IV.6 – YIELD TO MATURITY AND RETURN

- **Yield of a ZCB** with $C_T = 1$ at the maturity date T - The **annually compounded yield to maturity** ($n = 1$) is the value $Y_z(t, T)$ satisfying the equation:

$$P_z(t, T) = (1 + Y_z(t, T))^{-(T-t)},$$

and therefore we have $Y_z(t, T) = P_z(t, T)^{-1/(T-t)} - 1$;

- we call $Y_z(t, T)$ the annually compounded **zero-coupon yield**, or **zero-coupon rate**, or **spot rate** for date T .
- the function $(T - t) \mapsto Y_z(t, T)$ (for a given t) is called the (annually compounded) **zero-coupon yield curve** or simply the **yield curve**. It provides the same information as $(T - t) \mapsto P_z(t, T)$.
- The **continuously compounded yield to maturity** of the above mentioned coupon bond is the value $r_c(t, T)$ (also called *gross redemption yield*) satisfying the equation:

$$P_c(t, T) = \sum_{i=1}^n C_i \exp[-r_c(t, T) \times (T_i - t)].$$

IV.6 – YIELD TO MATURITY AND RETURN

- In the particular case of a ZCB, the **continuously compounded yield to maturity** $r_z(t, T)$ satisfies:

$$\begin{aligned} P_z(t, T) &= \exp[-r_z(t, T) \times (T - t)] \\ \Rightarrow r_z(t, T) &= -\frac{1}{T - t} \ln(P_z(t, T)) . \end{aligned} \quad (\text{IV.2})$$

- $r_z(t, T) = \ln(1 + Y_z(t, T))$, is also called the **term structure of interest rates**.
- There is an inverse relation between yield and associated bond price level.

IV.6 – YIELD TO MATURITY AND RETURN

- The **ZCB holding-period return** : is the return over some holding period less than the bond's residual maturity. Let us assume for ease of exposition that the holding period be a single period.
- Let us define $R_{t,t+1}^{T-t}$ the **arithmetic** one-period holding-period return on a ZCB purchased at time t at the price $P_z(t, T)$ and sold at time $t + 1$ at $P_z(t + 1, T)$. That is:

$$R_{t,t+1}^{T-t} = \frac{P_z(t+1, T)}{P_z(t, T)} - 1 \Rightarrow (1 + R_{t,t+1}^{T-t}) = \frac{(1 + Y_z(t, T))^{T-t}}{(1 + Y_z(t+1, T))^{T-t-1}}$$

- $R_{t,t+1}^{T-t}$ is HIGH if $Y_z(t, T)$ is HIGH or if $Y_z(t+1, T)$ is LOW.

IV.6 – YIELD TO MATURITY AND RETURN

- Let us take the **geometric** return $r_{t,t+1}^{T-t} = \ln(1 + R_{t,t+1}^{T-t})$, and we find:

$$\begin{aligned} & r_{t,t+1}^{T-t} \\ &= \ln\left(\frac{P_z(t+1, T)}{P_z(t, T)}\right) = (T-t)r_z(t, T) - (T-t-1)r_z(t+1, T) \\ &= r_z(t, T) - (T-t-1)(r_z(t+1, T) - r_z(t, T)) \end{aligned}$$

$r_{t,t+1}^{T-t}$ is determined by the beginning-of-period yield $r_z(t, T)$ (positively) and by the variation of the yield over the holding period (negatively).

- We also have that:

$$r_z(t, T) = \frac{1}{T-t} \sum_{i=0}^{T-t} r_{t+i,t+1+i}^{T-t-i}$$

and the continuously compounded yield on a ZCB equals the average continuously compounded return per period if the bond is held to maturity.

IV.6 – YIELD TO MATURITY AND RETURN

- **Spot EURIBOR rate** : the simply compounded interest rate convention is typically applied for commercial loans (fixed and floating reference rate loans). The most commonly used floating reference rate is the EURIBOR (London Interbank Offered Rate).
- Let us consider, at date t , the quoted EURIBOR annual rate with maturity date T , denoted $L(t, T)$ (say). The present value of 1 unit of money paid at T is:

$$P_L(t, T) = \frac{1}{[1 + L(t, T) \times (T - t)]},$$

and $P_L(t, T)$ can be seen as the **EURIBOR-based price** at date- t of a ZCB with unitary face value and maturing at T .

- Meaning: we deposit cash in EURIBOR-based deposits (like Eurodollar deposits with maturity longer than 1 week).
- Given that $C_T = 1$, we can identify $P_L(t, T)$ with the **EURIBOR-based Discount Factor** $Z_L(t, T)$.

IV.6 – YIELD TO MATURITY AND RETURN

- If in the market we have at date t a ZCB maturing at T with price $P_z(t, T)$, for arbitrage reasons we must have $P_L(t, T) = P_z(t, T)$. Indeed, let us imagine that, at time t , we have $P_L(t, T) > P_z(t, T)$. In that case, an investor can make a sure positive profit.
- He borrows at t , over the period $[t, T]$, the amount of money $P_L(t, T)$ at the simple rate $L(t, T)$ and he buys (again, at t) the ZCB paying the price $P_z(t, T)$.
- ⇒ At t he earns $P_L(t, T) - P_z(t, T)$, while at T he has a null profit : he obtain 1 unit of money from the ZCB and he provide the amount of money $P_L(t, T)[1 + L(t, T) \times (T - t)] = 1$.

IV.7 – FORWARD RATES AND DISCOUNT FACTORS

Definitions

- The annually compounded spot rate $r_1(t, T)$ set at date t concerns the price on a loan between the same date t (trading and settlement dates coincide) and the maturity date T .
- In the case of **forward rates**, the loan is received at some future settlement date $\tau \geq t$ and the maturity date is (as usual) $T > \tau \geq t$.
 - t = trading date,
 - τ = settlement date
 - and T = maturity date;
- We are fixing the rate of interest over the period $[\tau, T]$ in advance at time t . This is the rate which is appropriate at time t for discounting between τ and T .

IV.7 – FORWARD RATES AND DISCOUNT FACTORS

Definitions

- The **annually compounded forward rate**, denoted $f_1(t, \tau, T)$, with $t \leq \tau < T$, is the rate such that:

$$(1 + r_1(t, T))^{-(T-t)} = (1 + r_1(t, \tau))^{-(\tau-t)} \times (1 + f_1(t, \tau, T))^{-(T-\tau)}, \quad (\text{IV.3})$$

(when $t = \tau$ the forward rate reduces to the spot rate) and we deduce that:

$$f_1(t, \tau, T) = \frac{(1 + r_1(t, \tau))^{-(\tau-t)/(T-\tau)}}{(1 + r_1(t, T))^{-(T-t)/(T-\tau)}} - 1.$$

- we can also write (IV.3) in terms of discounting factors:

$$Z(t, T) = Z(t, \tau) \times (1 + f_1(t, \tau, T))^{-(T-\tau)}. \quad (\text{IV.4})$$

If relation (IV.3) (or (IV.4)) are not satisfied : arbitrage opportunity.

IV.7 – FORWARD RATES AND DISCOUNT FACTORS

Definitions

- The **short** (annually compounded) **forward rate** is given by:

$$f_1(t, \tau, \tau + 1) = \frac{Z(t, \tau)}{Z(t, \tau + 1)} - 1.$$

- This means that we can define the following relation between the discount factor and the sequence of short forward rates applying over time-to-maturity interval:

$$Z(t, \tau) = \prod_{j=t}^{\tau-1} \frac{Z(t, j+1)}{Z(t, j)} = \prod_{j=t}^{\tau-1} \frac{1}{1 + f_1(t, j, j+1)}.$$

IV.7 – FORWARD RATES AND DISCOUNT FACTORS

Definitions

- The **forward EURIBOR rate** at date t , valid for the period $[\tau, T]$, is the simply compounded rate $L(t, \tau, T)$ such that:

$$Z_L(t, \tau) = Z_L(t, T) \times [1 + L(t, \tau, T) \times (T - \tau)]. \quad (\text{IV.5})$$

This means that the forward rate implied by EURIBOR-based ZCBs $Z_L(t, \tau)$ and $Z_L(t, T)$ is:

$$L(t, \tau, T) = \frac{1}{T - \tau} \left(\frac{Z_L(t, \tau)}{Z_L(t, T)} - 1 \right).$$

- The **continuously compounded forward rate** prevailing at date t , for the period $[\tau, T]$, is the rate $f(t, \tau, T)$ such that:

$$\exp[-r(t, T) \times (T - t)] = \exp[-r(t, \tau) \times (\tau - t)] \times \exp[-f(t, \tau, T) \times (T - \tau)].$$

IV.7 – FORWARD RATES AND DISCOUNT FACTORS

Definitions

- This means that $f(t, \tau, T)$ is given by:

$$\begin{aligned} f(t, \tau, T) &= -\frac{1}{T - \tau} \ln \frac{Z(t, T)}{Z(t, \tau)}, \\ &= \frac{r(t, T)(T - t) - r(t, \tau)(\tau - t)}{T - \tau}. \end{aligned} \quad (\text{IV.6})$$

- the quantity $f(t, \tau, \tau + 1) = \ln \frac{Z(t, \tau)}{Z(t, \tau+1)}$ is called the **short forward rate** and gives the possibility to write:

$$Z(t, \tau) = \exp \left[- \sum_{j=t}^{\tau-1} f(t, j, j+1) \right]. \quad (\text{IV.7})$$

IV.7 – FORWARD RATES AND DISCOUNT FACTORS

Definitions

- We can also write (IV.3) in terms of **forward discount factor** $F(t, \tau, T)$:

$$F(t, \tau, T) = \frac{Z(t, T)}{Z(t, \tau)} = \frac{1}{(1 + f_1(t, \tau, T))^{(T-\tau)}}. \quad (\text{IV.8})$$

- The **forward discount factor** at time t defines the time value of money between two future dates, τ and $T > \tau$, and it is given by the ratio of the two date- t discount factors $Z(t, \tau)$ and $Z(t, T)$.
- We can equivalently write the forward discount factor as:

$$F(t, \tau, T) = \frac{P_z(t, T)}{P_z(t, \tau)}, \quad (\text{IV.9})$$

where the two ZCBs have the same face value (e.g. \$100).

- The forward discount factor has the following properties: *i*) $F(t, \tau, T) = 1$ for $T = \tau$; *ii*) $F(t, \tau, T)$ is decreasing in T .

IV.7 – FORWARD RATES AND DISCOUNT FACTORS

Definitions

- The **forward rate** at time t for a risk-free investment from τ to T , and **with compounding frequency n** , is the interest rate determined by $F(t, \tau, T)$:

$$f_n(t, \tau, T) = n \times \left(\frac{1}{F(t, \tau, T)^{\frac{1}{n(T-\tau)}}} - 1 \right).$$

- The **continuously compounded forward rate** is obtained for $n \rightarrow +\infty$:

$$f(t, \tau, T) = -\frac{1}{(T - \tau)} \ln(F(t, \tau, T)).$$

- Given an n -times compounded forward rate $f_n(t, \tau, T)$, the discount factor is:

$$F(t, \tau, T) = \frac{1}{\left(1 + \frac{f_n(t, \tau, T)}{n}\right)^{n(T-\tau)}}.$$

IV.7 – FORWARD RATES AND DISCOUNT FACTORS

The Forward Curve

- Given a continuously compounded forward rate $f(t, \tau, T)$, we have:

$$F(t, \tau, T) = \exp(-f(t, \tau, T)(T - \tau)).$$

If the discount factor $Z(t, T)$ is increasing between two dates τ and $T > \tau$, that is $Z(t, \tau) < Z(t, T)$, then the forward rate at t for an investment between τ and T is negative. Nevertheless, $Z(t, T)$ is decreasing in T .

- Consider the following continuously compounded rate:

$$f(t, \tau, \tau + \Delta) = -\frac{\ln(F(t, \tau, \tau + \Delta))}{\Delta}$$

The forward curve gives the relation between the forward rate $f(t, \tau, \tau + \Delta)$ and the time of the investment τ (for a fixed Δ).

- Forward curve is related, to some extent, to the market expectations of future nominal interest rates, as well as market participants degree of aversion to risky investments.

IV.7 – FORWARD RATES AND DISCOUNT FACTORS

The Forward Curve

- The Forward curve is derived from the same discount factors that determine the Spot Curve.
- Indeed, we have that:

$$\begin{aligned}f(t, \tau, T) &= -\frac{1}{T - \tau} \ln \frac{Z(t, T)}{Z(t, \tau)} = \frac{r(t, T)(T - t) - r(t, \tau)(\tau - t)}{T - \tau} \\&= r(t, \tau) + (T - t) \frac{r(t, T) - r(t, \tau)}{T - \tau}.\end{aligned}\tag{IV.10}$$

- We find that, for any maturity τ , the forward rate between τ and T equals the spot rate $r(t, \tau)$ plus a term that is positive if the spot curve is rising, and it is negative if the spot curve is declining at τ . Thus, when the spot curve is:
 - i) **increasing** → the forward curve is **above** the spot curve.
 - ii) **decreasing** → the forward curve is **below** the spot curve.
 - iii) **flat** → the forward curve **equals** the spot curve.

IV.7 – FORWARD RATES AND DISCOUNT FACTORS

The Forward Curve

Figure 5.2 The Spot Curve and the Forward Curve

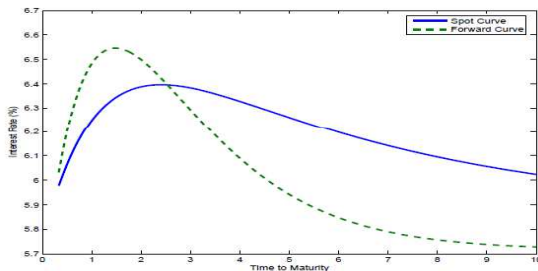


Figure 6: From Veronesi (2010, Chapter 5).

IV.8 – EXTRACTING YIELD CURVES FROM BOND PRICES

Introduction

- In general only a few ZCBs are traded, so that information about $Z(t, T)$ must be inferred from $P_c(t, T)$.
 - We present here methods to extract / estimate the term structure of interest rates from $P_c(t, T)$. We assume that $Z(t, T)$ is specified as a function of unknown parameters (and the time-to-maturity $T - t$):
 $Z(t, T) = f(t, T, \theta)$:
 - a) the vector of parameters θ is estimated in order to obtain the best possible fit (for a given estimation criterion) of observed $P_c(t, T)$ by theoretical ones.
 - b) $f(t, T, \theta)$ = exponential function of $\tau = T - t$ (for a given t). This is consistent with the idea that $Z(t, T)$ and $r(t, T)$ are continuous and smooth function of τ . Indeed, if $r(t, T_1) \gg r(t, T_1 + \varepsilon)$:
- ⇒ Bond owners would probably shift from low-yield to high-yield bonds, and bond issuers would shift to the low-yield maturity. These changes in the supply and demand will determine a reduction of the gap driving $r(t, T_1)$ close to $r(t, T_1 + \varepsilon)$.

IV.8 – EXTRACTING YIELD CURVES FROM BOND PRICES

Exponential-Polynomial Families: Nelson and Siegel (1987)

- The purpose of this section is to introduce the parametric curve families which are used by most central banks to construct the term structure of interest rates.
- The main feature is to capture (to fit) as much as possible the structure of market interest rates with a small number of parameters. The Nelson and Siegel (1987) yield curve is parameterized as :

$$\begin{aligned}r(t, t+h) &= \beta_0 + \beta_1 \frac{1 - e^{(-h/\theta)}}{h/\theta} + \beta_2 \frac{1 - \left(1 + \frac{h}{\theta}\right) e^{(-h/\theta)}}{h/\theta} \\&= \beta_0 \times B_0(h) + \beta_1 \times B_1(h) + \beta_2 \times B_2(h)\end{aligned}$$

where $B_0(h) = 1$,

$$B_1(h) = \frac{1 - e^{(-h/\theta)}}{h/\theta},$$

$$B_2(h) = \frac{1 - \left(1 + \frac{h}{\theta}\right) e^{(-h/\theta)}}{h/\theta}.$$

IV.8 – EXTRACTING YIELD CURVES FROM BOND PRICES

Exponential-Polynomial Families: Nelson and Siegel (1987)

□ Interpretation of the parameters:

θ = scale (or "hump" location) parameter ,

β_0 = level parameter ,

β_1 = slope parameter , β_2 = curvature parameter ,

□ Indeed

(i) if $h \rightarrow +\infty$ (and $\theta > 0$) then $r(t, t+h) \rightarrow \beta_0$, (long-term yield)

(ii) if $h \rightarrow 0$ then $r(t, t+h) \rightarrow (\beta_0 + \beta_1)$ = (very) short rate

$\Rightarrow \beta_1$ it drives the yield curve spread between .

IV.8 – EXTRACTING YIELD CURVES FROM BOND PRICES

Exponential-Polynomial Families: Nelson and Siegel (1987)

Econometrics of
Commodity and
Asset Pricing

Lecture 2

Monfort &
Pegoraro & Renne

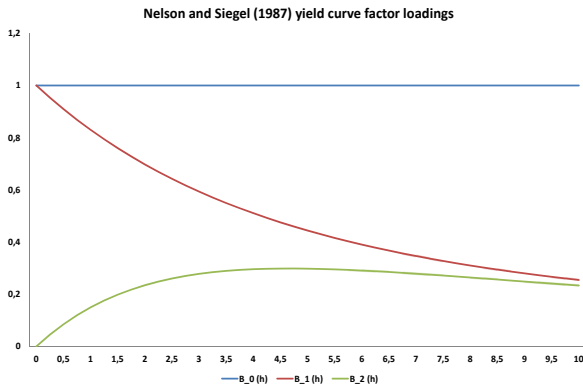


Figure 7: Nelson and Siegel (1987) yield curve loadings ($\theta = 2.6$).

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IV.8 – EXTRACTING YIELD CURVES FROM BOND PRICES

Exponential-Polynomial Families: Nelson and Siegel (1987)

- The discount function $Z(t, t + h)$ (i.e., the ZCB price with unitary face value) is:

$$Z(t, t + h) = \exp \left[-a h - b \theta (1 - e^{(-h / \theta)}) - c h e^{(-h / \theta)} \right] .$$

where $a = \beta_0$, $b = \beta_1 + \beta_2$ and $c = -\beta_2$.

- We write, for any $i \in \{1, \dots, M\}$, the following regression:

$$\begin{aligned} & P_{c,i}(t, t + h_i) \\ &= \sum_{j=1}^m C_{i,j} \left\{ \exp \left[-a h_i - b \theta (1 - e^{(-h_i / \theta)}) - c h_i e^{(-h_i / \theta)} \right] \right\} + \varepsilon_t^{(i)} . \end{aligned}$$

→ and parameters are estimated by non-linear regression techniques [see Gallant (1987)].

IV.8 – EXTRACTING YIELD CURVES FROM BOND PRICES

Exponential-Polynomial Families: Svensson (1994)

□ Svensson (1994) generalizes the Nelson-Siegel curve adding a third term.

□ Precisely, he assumes:

$$r(t, t+h) = \beta_0 + \beta_1 \frac{1 - e^{-h/\theta_1}}{h/\theta_1} + \beta_2 \frac{1 - \left(1 + \frac{h}{\theta_1}\right) e^{-h/\theta_1}}{h/\theta_1} \\ + \beta_3 \frac{1 - \left(1 + \frac{h}{\theta_2}\right) e^{-h/\theta_2}}{h/\theta_2}.$$

□ The new parameters β_3 and θ_2 provide additional flexibility, compared with the Nelson-Siegel curve, and in particular over medium- to long-term maturities.

□ In particular, θ_2 allows for a second "hump" in the yield curve.

IV.8 – EXTRACTING YIELD CURVES FROM BOND PRICES

Exponential-Polynomial Families: Svensson (1994)

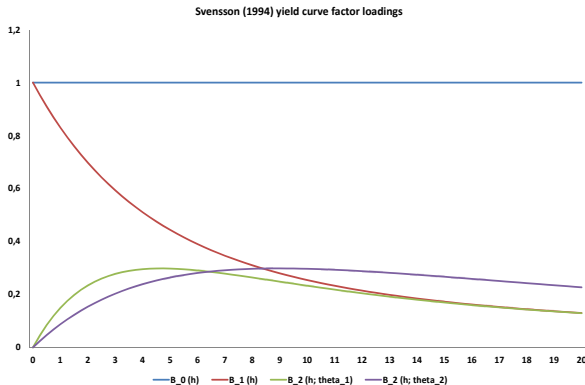


Figure 8: Svensson (1994) yield curve loadings.

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Exponential-Polynomial Families: Svensson (1994)

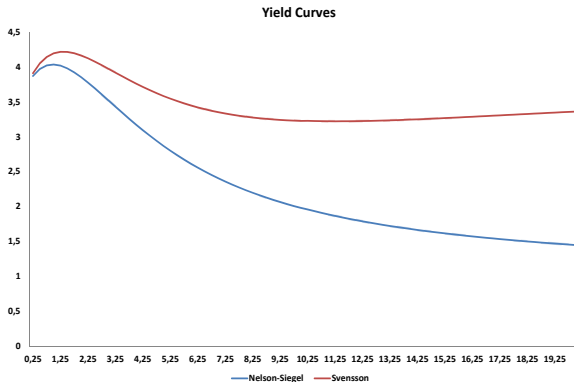


Figure 9: Nelson-Siegel and Svensson yield curves.

IV.8 – EXTRACTING YIELD CURVES FROM BOND PRICES

The Gurkaynak, Sack and Wright (2007) data base

- Among the several U.S. yield curve data bases characterizing the empirical studies in the fixed income literature [Fama and Bliss (1987), McCulloch and Kwon (1993)], Gurkaynak, Sack and Wright (2007) have proposed daily U.S. yield curve estimates.
- The sample period is from 1961 to the present: it is updated regularly and is available (for free!) at
<http://www.federalreserve.gov/Pubs/feds/2006/200628/200628abs.html>
<http://www.federalreserve.gov/econresdata/researchdata.htm>
- The methodology they use to estimate (daily!) the forward curve $f(t, t + h)$ is using the Svensson (1994) family, that is a generalization of the Nelson and Siegel (1987) family.
- The estimation is based on observations of only T-notes and T-bonds prices (maturities, at the issuing date, ranging from 2 years to 30 years).

IV.8 – EXTRACTING YIELD CURVES FROM BOND PRICES

The Gurkaynak, Sack and Wright (2007) data base

- At any date t in the sample, they thus exclude:
 - all T-bills : their market is disconnected (segmented) from that of T-notes and T-bonds.
 - all T-notes and T-bonds with residual maturity less than 3 months : they are highly affected by lack of liquidity of short-term investors.
 - the two most recently issued T-notes and T-bonds: they are called "on-the-run" and "first off-the-run". They are excluded because of their greater liquidity.
- The bonds have to :
 - be denominated in the same currency,
 - be of the same credit quality (default-free in our case!),
 - reflect the same level of liquidity,
 - and have not be affected by option like features (callable bonds).
- Ideally, the bonds should ideally differ in terms of their maturity and coupons.

IV.8 – EXTRACTING YIELD CURVES FROM BOND PRICES

The Gurkaynak, Sack and Wright (2007) data base

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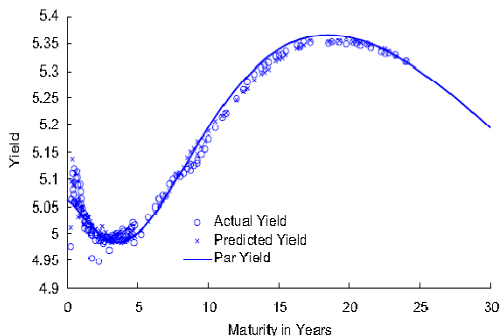


Fig. 1. Par-yield curve on May 9, 2006.

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Figure 5: Decomposition of the Yield Curve on May 3, 2006

